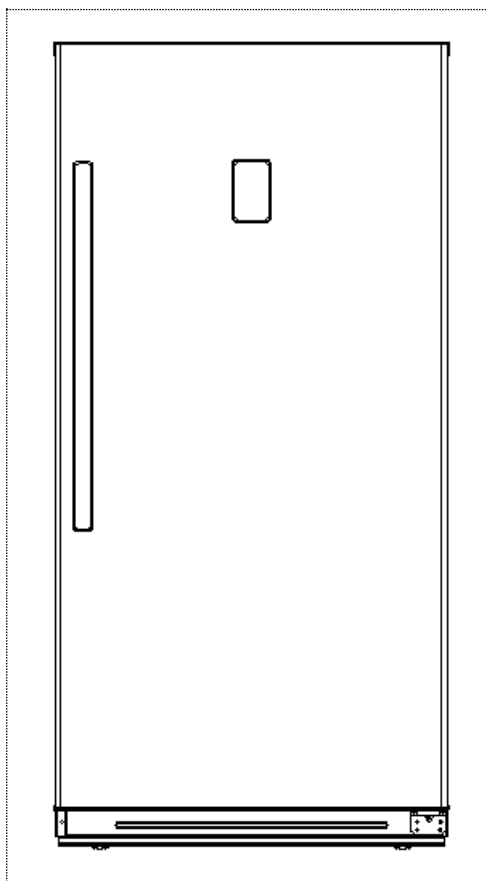


INSIGNIA®

SERVICE MANUAL

Upright Frz. Series

Applicable Models	Product Models
HS-625FWEN	NS-UZ17WH0
HS-625FWEN	NS-UZ17SS0
HS-772FWE	NS-UZ21WH0



The picture in this service manual is only for reference, and specific appearance and configuration are subject to the real product.

This manual mainly teaches the method, the specific work skill needs engineer to accumulate through the daily work.

 WARNING

Important Safety Notice

There are special components used in this equipment which are important for safety. These parts are marked by  in the Schematic Diagrams, Circuit Board Diagrams, Exploded Views and Replacement Parts List. It is essential that these critical parts should be replaced with manufacturer's specified parts to prevent shock, fire or other hazards. Do not modify the original design without permission of manufacturer.

 WARNING

Important Safety Notice

The Maintenance Manual is only for the use of maintenance personnel with certain experience and background in electrical, electronic and mechanical field.

Any attempt to repair main devices may lead to personal injury and property loss.

Manufacturers or distributors are not responsible for the content of the Manual and interpretation thereof.

Contents

1. SIGNIFICANT UPDATE NOTES (NONE)	4
2. SAFETY WARNING	5
2.1 WARNING FOR OPERATION SAFETY	5
2.2 SAFETY INSTRUCTION FOR REFRIGERANT	7
3. INSTALLATION AND COMMISSIONING	8
3.1 HANDLING	8
3.2 DOOR DISASSEMBLY AND ASSEMBLY	8
3.3 INSTALLATION LOCATION	8
3.4 LEVELING OF THE REFRIGERATOR	9
3.5 LEFT OR RIGHT OPEN DOOR REVERSAL (NONE)	9
3.6 INSTALLATION OF HANDLE	9
3.7 INSTALLATION OF DOOR LOCK (NONE)	9
4. MAIN PARTS AND EXTERNAL DIMENSION	10
4.1 MAIN PARTS	10
4.2 EXTERNAL DIMENSION	11
4.3 MIDEA PRODUCT SERIAL NUMBER AND LOCATION	12
5. ELECTRIC CONTROL SYSTEM	13
5.1 ELECTRICAL PARTS PARAMETERS	13
5.2 CIRCUIT DIAGRAM	15
5.3 MAIN PCB TERMINAL CONNECTION DIAGRAM	16
5.4 INVERTER BOARD TERMINAL CONNECTION DIAGRAM (NONE)	16
6. REFRIGERATION SYSTEM	17
6.1 REFRIGERATION SYSTEM WORKING PRINCIPLE	17
6.2 COOLING PIPELINE AND DRAIN PIPE INSIDE THE CABINET	17
6.3 CIRCULATING ROUTE OF COOLING AIR	18
6.4 WELDING POINTS IN CHAMBERS OR FOAM LAYER	18
6.5 WELDING POINT IN THE COMPRESSOR CASE	19
7. DISMANTLING OF PARTS	20
7.1 PARTS ON THE DOOR	20
7.2 PARTS INSIDE THE REFRIGERATOR	20
7.3 LIGHT SYSTEM	21
7.4 AIR DUCT COMPONENTS IN REFRIGERATING CHAMBER AND FAN MOTOR (NONE)	21
7.5 AIR DUCT COMPONENTS IN FREEZING CHAMBER AND FAN MOTOR	22
7.6 EVAPORATOR AND DEFROST SYSTEM	23
7.7 COMPRESSOR CASE	24
7.8 DISPLAY CONTROL BOARD	28
7.9 MAIN CONTROL BOARD	29

7.10 VARIABLE FREQUENCY DRIVER BOARD (NONE)	30
7.11 WATER DISPENSER (NONE)	30
8. TEMPERATURE SENSING SYSTEM.....	31
8.1 POSITION OF SENSORS	31
8.2 REPLACEMENT OF SENSORS	31
8.3 SENSOR WITHOUT TERMINAL REPLACEMENT	33
8.4 SENSOR R/T TABLE.....	34
9. FUNCTION AND OPERATION.....	35
9.1 DISPLAY OPERATION PANEL	35
9.2 TEMPERATURE CONTROL	36
9.3 MODE SETTING.....	36
9.4 DEFROSTING FUNCTION	37
9.5 OPEN DOOR ALARM.....	37
9.6 ERROR CODE AND SOLUTIONS	38
9.7 TEST MODE.....	40
9.8 DEMO MODE	41
9.9 BACKUP DATA FOR POWER FAIL	41
10. COMPRESSOR.....	42
10.1 COMPRESSOR ON AND OFF CONTROL SPECIFICATIONS	42
10.2 INVERTER BOARD FAULT ANALYSIS (NONE)	42
11. TROUBLESHOOTING METHOD.....	43
11.1 NO COOLING (AIR COOLING-ELECTRONIC).....	43
11.2 NO WORKING OF COMPRESSOR.....	45
11.3 INSIDE FROSTING, NO DEFROSTING	46
11.4 INSIDE FROSTING, NO DEFROSTING-MAINTENANCE GUIDELINES	47
11.5 LIGHT IS NOT ON	48
11.6 FAN FAILURE.....	49
11.7 DEFECTIVE DEFROST CIRCUIT	49
11.8 NOISE	50
11.9 AIR DUCT NOT OPERATED (OPTION)	50
11.10 ICE MAKER NOT MAKE ICE (OPTION)	51
12. PRODUCT EXPLODED VIEW AND SPARE PARTS LIST	ERROR! BOOKMARK NOT DEFINED.

1. Significant update notes (None)

2. Safety Warning

2.1 Warning for operation safety

Important Safety Instructions

	CAUTION RISK OF ELECTRIC SHOCK DO NOT OPEN	
	This symbol indicates that dangerous voltage constituting a risk of electric shock is present within your freezer.	
	This symbol indicates that there are important operating and maintenance instructions in the literature accompanying your freezer.	

WARNING

- 1) Read these instructions.
- 2) Keep these instructions.
- 3) Heed all warnings.
- 4) Follow all instructions.
- 5) Do not use this appliance near water.
- 6) Clean only with a damp cloth.
- 7) Do not block any ventilation openings.
- 8) Install in accordance with the manufacturer's instructions.
- 9) Do not install near any heat sources, such as radiators, heat registers, stoves, or other apparatus that produce heat.
- 10) Do not defeat the safety purpose of the polarized or grounding-type plug. A polarized plug has two blades with one wider than the other. A grounding type plug has two blades and a third grounding prong. The wide blade or the third prong are provided for your safety. If the provided plug does not fit into your outlet, consult an electrician for replacement of the obsolete outlet.
- 11) Protect the power cord from being walked on or pinched, particularly at plugs, convenience receptacles, and the point where they exit from the appliance.
- 12) Do not attempt to modify or extend the power cord of this appliance.
- 13) Unplug this appliance during lightning storms or when it will not be used for long periods of time.
- 14) Make sure that the available AC power matches the voltage requirements of this appliance.

CONNECTING ELECTRICITY

**WARNING****Electrical Shock Hazard.**

Plug into a grounded 3-prong outlet.
Do not remove the ground prong.
Do not use an adapter.

Failure to follow these instructions can result in death, fire, or electrical shock.

**WARNING****Electric Shock Hazard**

Failure to follow these instructions can result in electric shock, fire, or death.

- 1) **WARNING**—Keep ventilation openings, in both the freezer and the built-in structure, clear of obstruction.
- 2) **WARNING**—Do not touch the interior of the freezer with wet hands. This could result in frost bite.
- 3) **WARNING**—Do not use mechanical devices or other means to accelerate the defrosting process, other than those recommended by the manufacturer.
- 4) **WARNING**—Do not damage the refrigerant circuit.
- 5) **WARNING**—Do not damage the refrigerant tubing when handling, moving, or using the freezer.
- 6) **WARNING—DANGER**—Never allow children to play with, operate, or crawl inside the freezer. Risk of child entrapment. Before you throw away your old freezer:
 - 6-1) Take off the doors
 - 6-2) Leave the shelves in place so that children may not easily climb inside
- 7) Unplug the freezer before carrying out user maintenance on it.
- 8) This freezer can be used by children age eight years and older and persons with reduced physical or mental capabilities or lack of experience and knowledge if they are given supervision or instruction concerning the use of the freezer in a safe way and understand the hazards involved. Children should not play with the freezer. Cleaning and maintenance should not be performed by children without supervision.
- 9) If a component part is damaged, it must be replaced by the manufacturer, its service agent, or similar qualified persons in order to avoid a hazard.
- 10) Please dispose of the freezer according to local regulations as the freezer contains flammable gas and refrigerant.
- 11) Follow local regulations regarding disposal of the freezer due to flammable refrigerant and gas. All refrigeration products contain refrigerants, which under the guidelines of federal law must be removed before disposal. It is the consumer's responsibility to comply with federal and local regulations when disposing of this product.
- 12) This freezer is intended to be used in household and similar environments.

13) Do not store or use gasoline or any flammable liquids inside or in the vicinity of this freezer.

14) Do not use extension cords or ungrounded (two-prong) adapters with this freezer. If the power cord is too short, have a qualified electrician install an outlet near the freezer. Use of an extension cord can negatively affect the freezer's performance.

Grounding requirement

This freezer must be grounded. This freezer is equipped with a cord having a grounding wire with a grounding plug. The plug must be inserted into an outlet that is properly installed and grounded.

Improper use of the grounding plug can result in a risk of electric shock. Consult a qualified electrician or service person if the grounding instructions are not completely understood, or if doubt exists as to whether the freezer is properly grounded.

2.2 Safety instruction for refrigerant



WARNING



Explosion Hazard.

Keep flammable materials and vapors, such as gasoline, away from freezer. Failure to do so can result in fire, explosion, or death.



Safety instruction for refrigerant

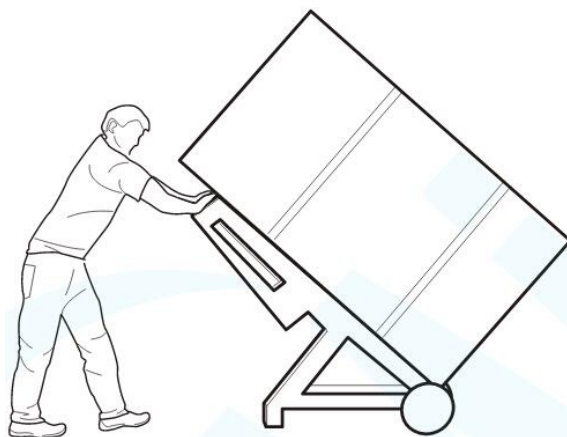
DANGER–Risk of Fire or Explosion. Flammable Refrigerant Used. To Be Repaired Only By Trained Service Personnel. Do Not Use Mechanical Devices. Do Not Puncture Refrigerant Tubing. CAUTION–Risk of Fire or Explosion. Flammable Refrigerant Used. Consult Repair Manual/Owner's Guide Before Attempting To Service This Product. All Safety Precautions Must be Followed. CAUTION–Risk of Fire or Explosion. Dispose of Properly In Accordance With Federal Or Local Regulations. Flammable Refrigerant Used. CAUTION–Risk of Fire or Explosion Due To Puncture Of Refrigerant Tubing; Follow Handling Instructions Carefully. Flammable Refrigerant Used.

3. Installation and commissioning

3.1 Handling

Handling

- 1) Protect the refrigerator in moving it, Same as shown as left photo, please move it by handcart with cushion
- 2) Remove all packing materials and bottom cushion, the move into house for placement
- 3) After moving it to appropriate location, wait for 2 hours before power on.



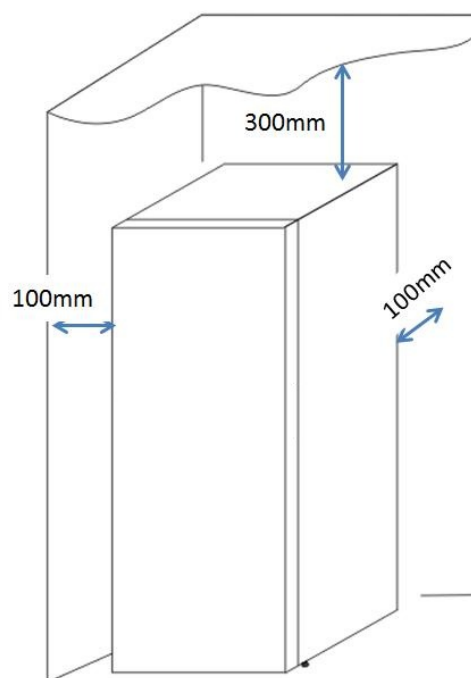
3.2 Door Disassembly and Assembly

When the whole refrigerator cannot enter the room, the door can be disassembled, then assembled after entering separately.

3.3 Installation location

Installation location

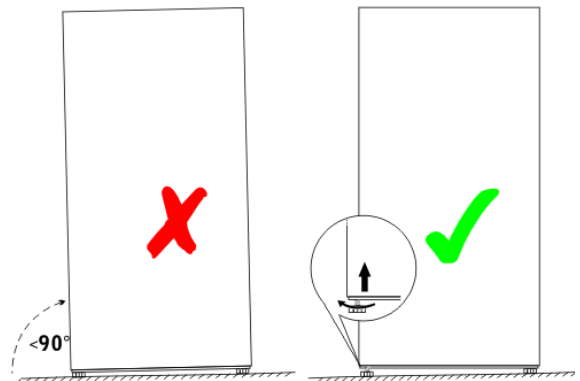
Please select a ventilated place to place the refrigerator, and reserve space according to the recommended size in the picture, which is conducive to heat dissipation, performance improvement and energy consumption reduction.



3.4 Leveling of the refrigerator

Leveling of the refrigerator

If the refrigerator cannot be placed steadily, adjust the footing to level it. Turn the feet clockwise to raise the refrigerator; turn the feet counterclockwise to lower the refrigerator.

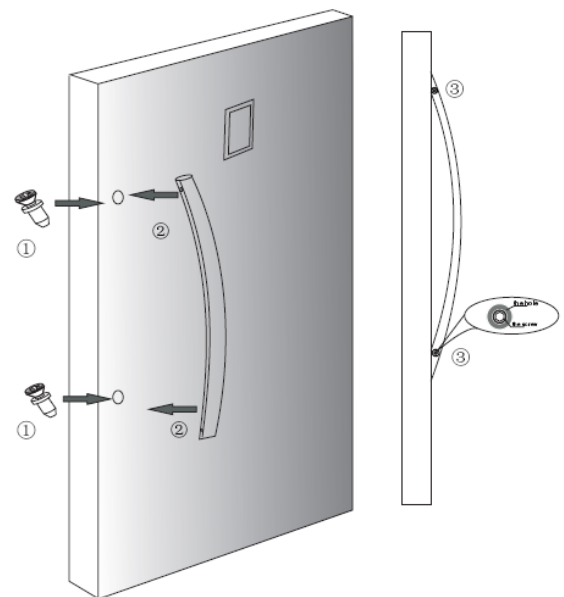


3.5 Left or right open door reversal (None)

3.6 Installation of handle

Installation of handle

- 1) Take two handle screws, install them in the door hole
 - 2) Alignment screw installation handle
 - 3) Fix screws in the side hole of the handle.
- Note: The wrench and screws are placed in the appendage bag.

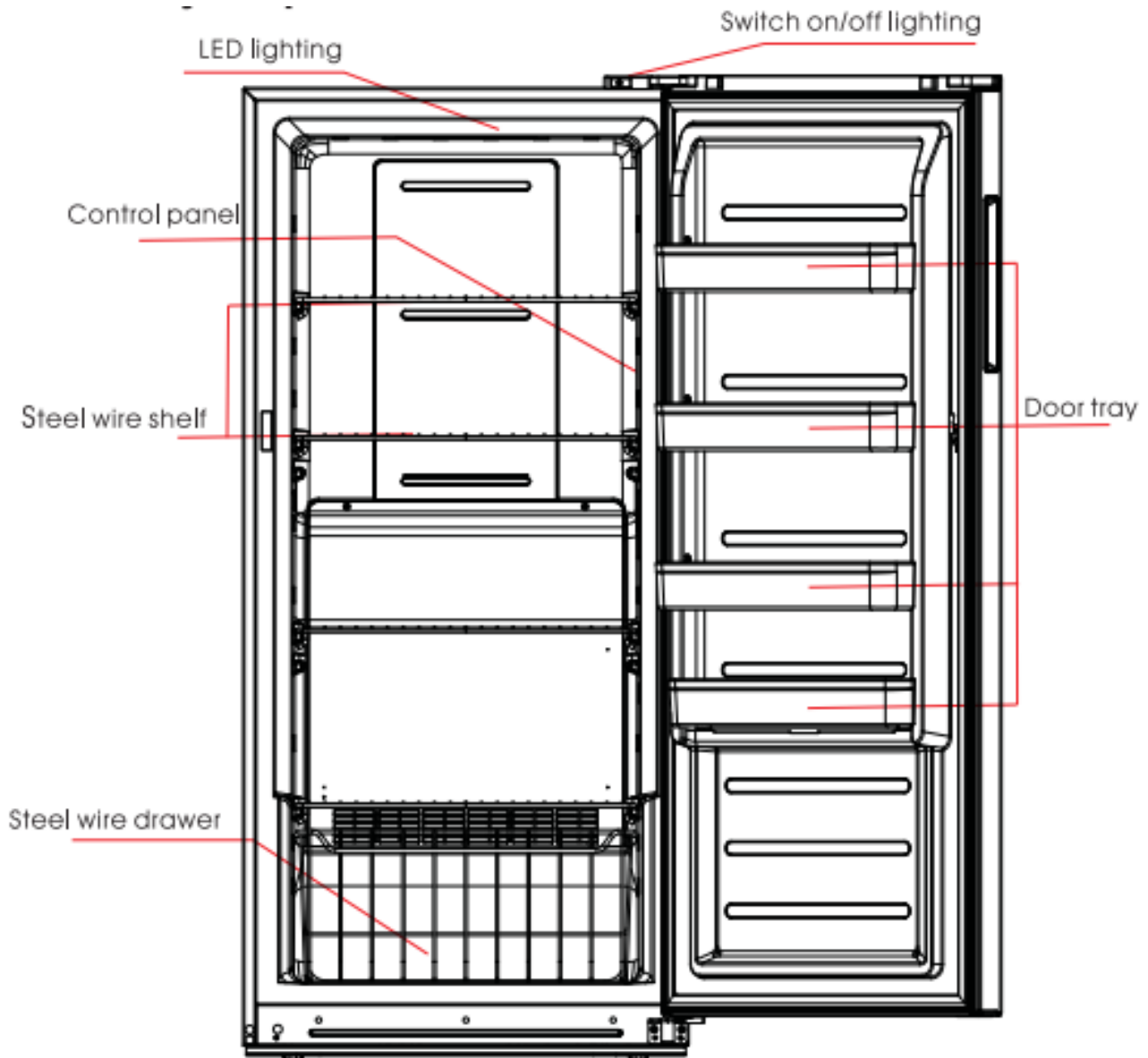


3.7 Installation of door lock (None)

4. Main parts and external dimension

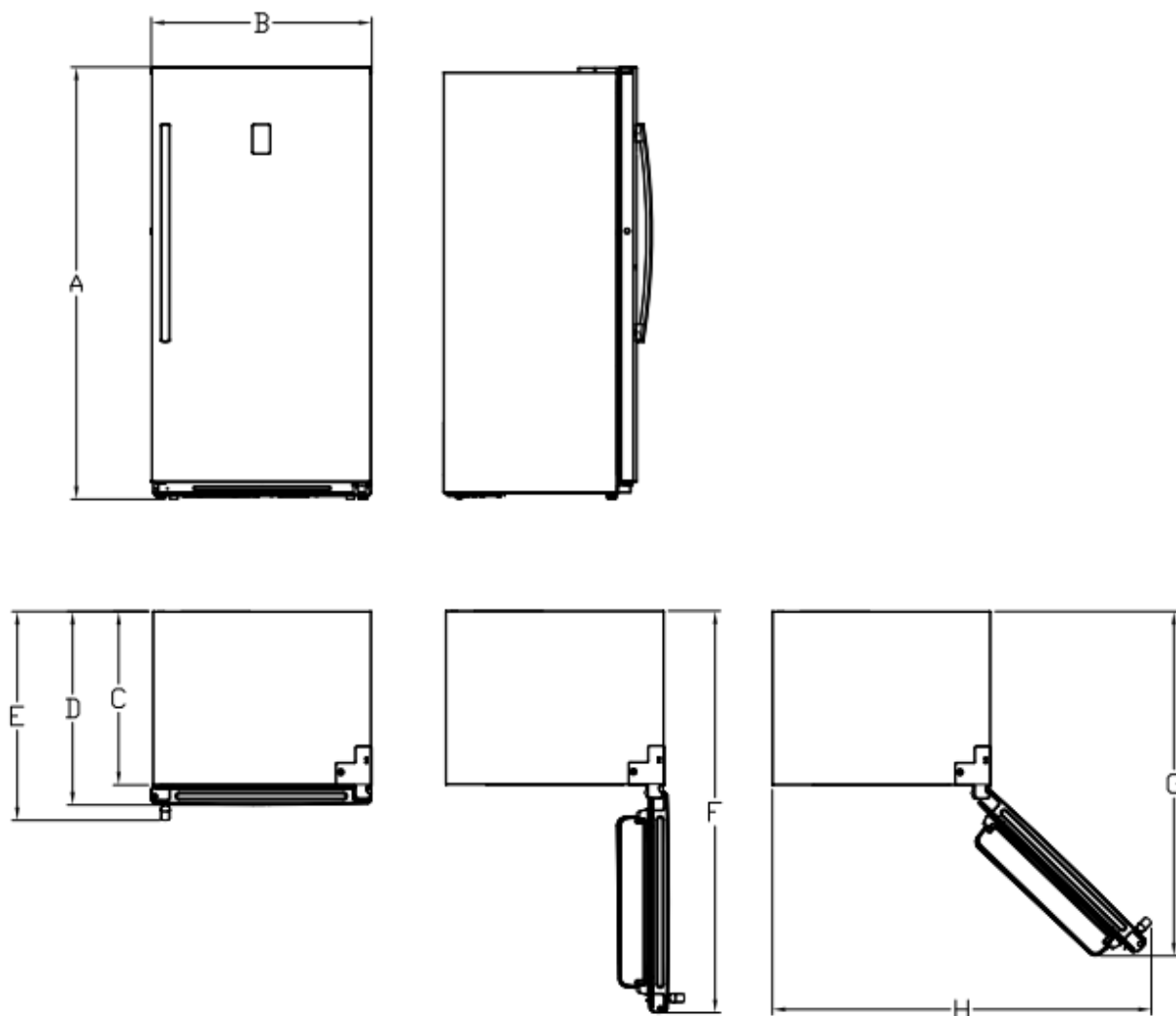
4.1 Main parts

(The picture is only for reference, and specific appearance and configuration are subject to the real product)



4.2 External dimension

Description	Code	UZ17 Size (mm)	UZ21 Size (mm)
Height to top of Cabinet	A	1650	1950
Width	B	830	830
Depth w/Cabinet	C	660	660
Depth w/Door	D	735	735
Depth w/Handle	E	794	794
Depth (Door open 90 deg. w)	F	1528	1528
Depth (Door open 135 deg. w)	G	1310	1310
Width (Door open 135 deg. w)	H	1442	1442



(The picture is only for reference)

4.3 Midea product serial number and location

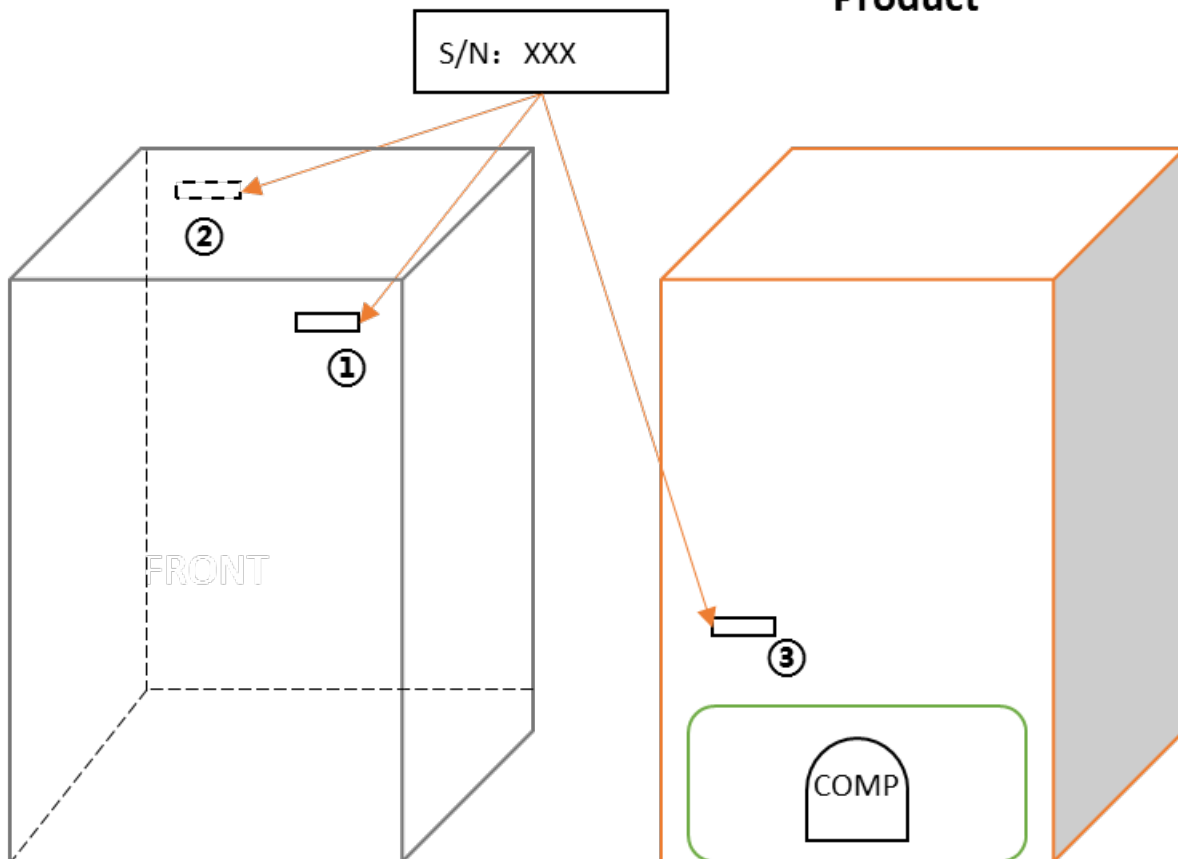
1) **Product Serial Number** — Including order number, production date and other information. When the product occur problem, it needs to be recorded or photographed and provided to us.



2) Paste location

Carton Box

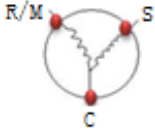
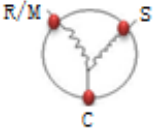
Product



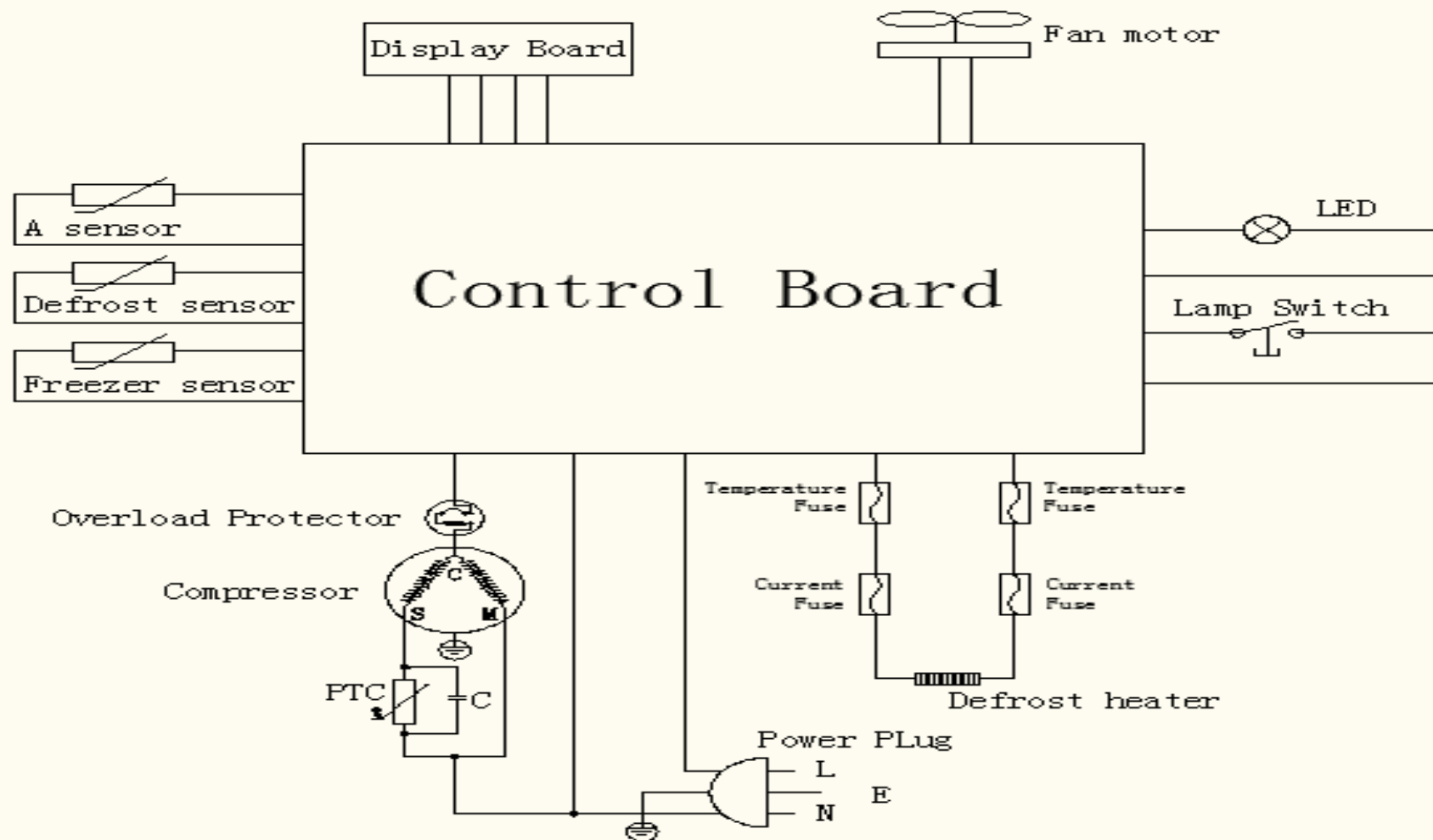
Some products also have S/N on the lower part of the right side of the Cabinet.

5. Electric control system

5.1 Electrical parts parameters

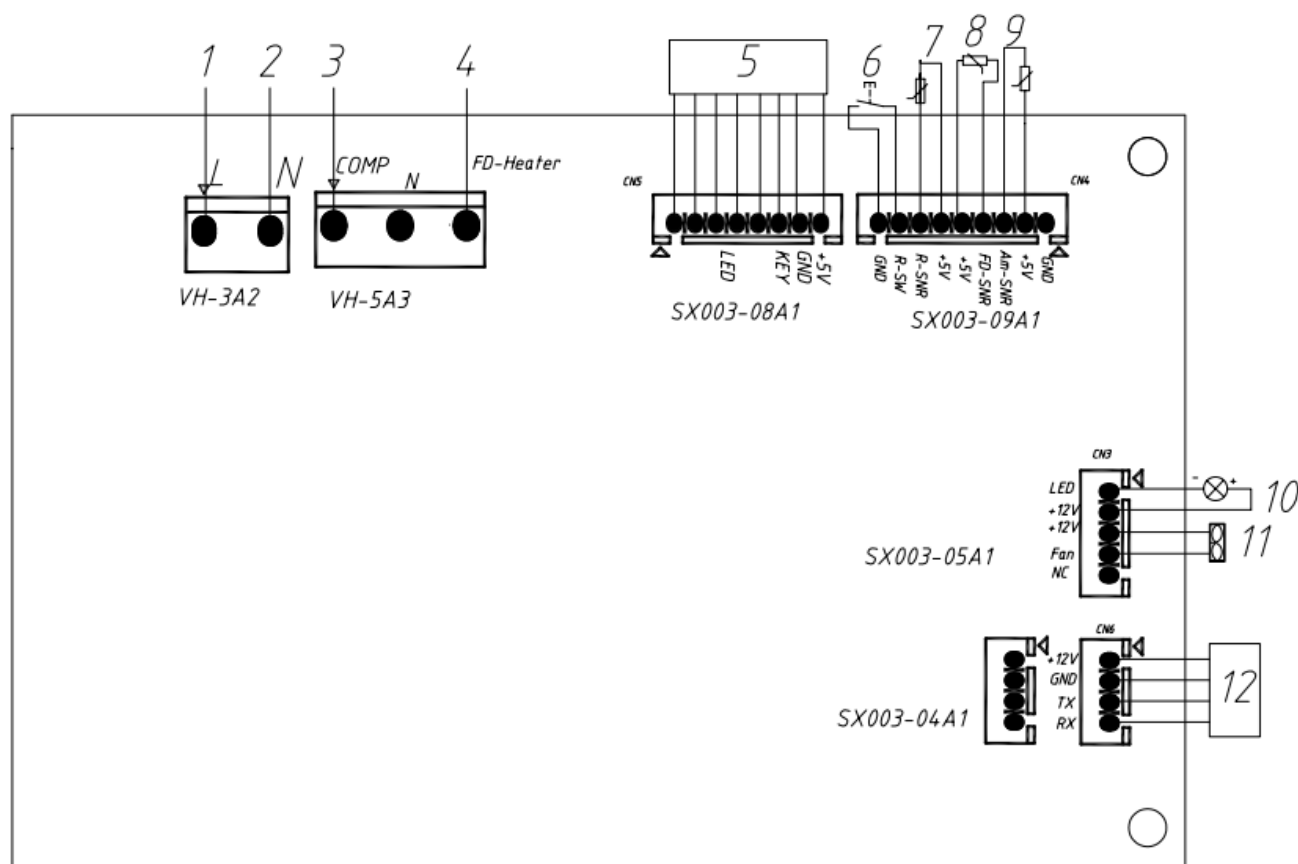
Applicable Model	HS-774FWEN	HS-774FWEN
Product Model	NS-UZ17	NS-UZ21
Product code	22031010002925	22031010002926
Rated Voltage	115V、60Hz	115V、60Hz
Item	Specification	Specification
Refrigerant	R600a	R600a
Compressor	EZ75E1A (GMCC) (Part code : 11101010011093)	EZ90H1A-U (GMCC) (Part code : 11101010011794)
Starting device type	Fixed Speed	Fixed Speed
The COP of compressor	1.64 (W/W)	1.70 (W/W)
The max cooling capacity of compressor	153 W	190 W
Winding resistance of compressor wiring terminal (20°C)	Rmc:5.35Ω±7% Rsc:8.01Ω±7% Rms = Rmc + Rsc	Rmc:4.03Ω±7% Rsc:4.21Ω±7% Rms = Rmc + Rsc
Winding resistance picture		
Starter(PTC)	8EA14C3-02 (Part code : 12031000001063)	8EA14C3-02
Overload protector(OLP)	4TM302TFBYY (Part code : 12031000001311)	4TM3 19TFBYY
Integrate PTC+OLP	None	None
Variable frequency driver board	None	None
Capacitor	8μF、250V	12μF、250V
Power filter (EMI)	None	None
Power reactor (EU EMC)	None	None
Motor		
Fan motor of the freezing chamber	DC12V、2.4 W、2020r/min	DC12V、2.4 W、2020r/min
Fan motor of the refrigerating chamber	None	None
Electric damper	None	None
Lights		
Lights inside the top of refrigerating chamber	None	None
Lights inside the freezing chamber	DC12V、≤6W	DC12V、≤6W

Others Lights	None	None
Switch of the light	☒ Mechanical switch ☐ Magnetism control switch	☒ Mechanical switch ☐ Magnetism control switch
Defrosting parts		
Defrosting sensor	NTC B3839 (B5/25=3839K±2%)	
Fuse in freezing chamber	115V、77(0 ~ -5) °C	
Defrost heater in freezing chamber	115V、320W	

5.2 Circuit diagram

F: Freezer R: Refrigerator A: Ambient temperature

5.3 Main PCB terminal connection diagram



The connecting terminals list

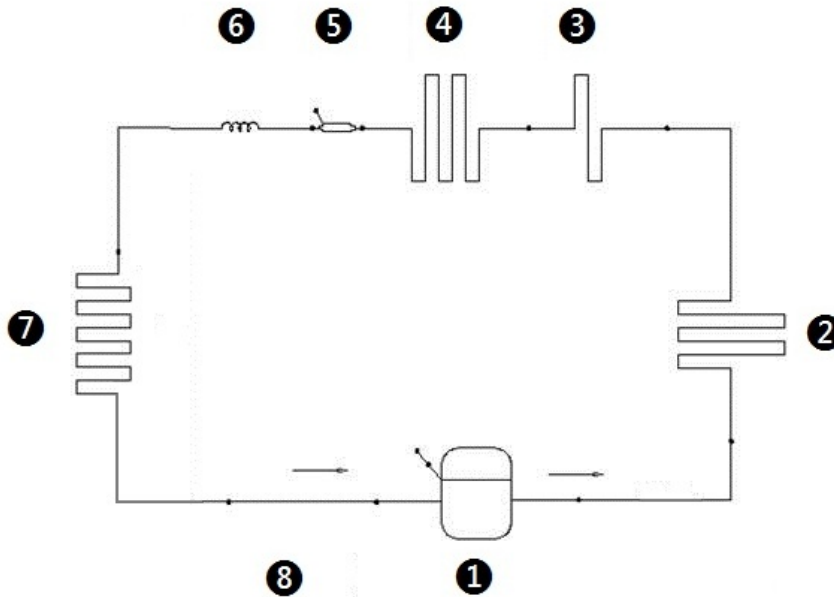
Connecting terminals	Connecting terminals
1+2.Power supply	8. Freezing defrost sensor
3. Compressor	9. Ambient temperature sensor
4. Freezing defrost heater	10. LED of freezing chamber
5.Display control panel (Internal)	11. Freezing fan motor
6. Light switch	12. Display control panel(outer)
7. Freezing temperature sensor	

5.4 Inverter board terminal connection diagram (None)

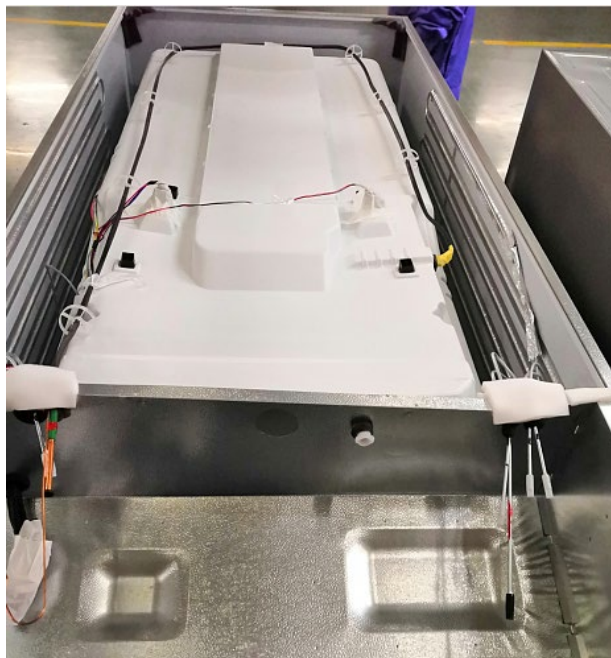
6. Refrigeration system

6.1 Refrigeration system working principle

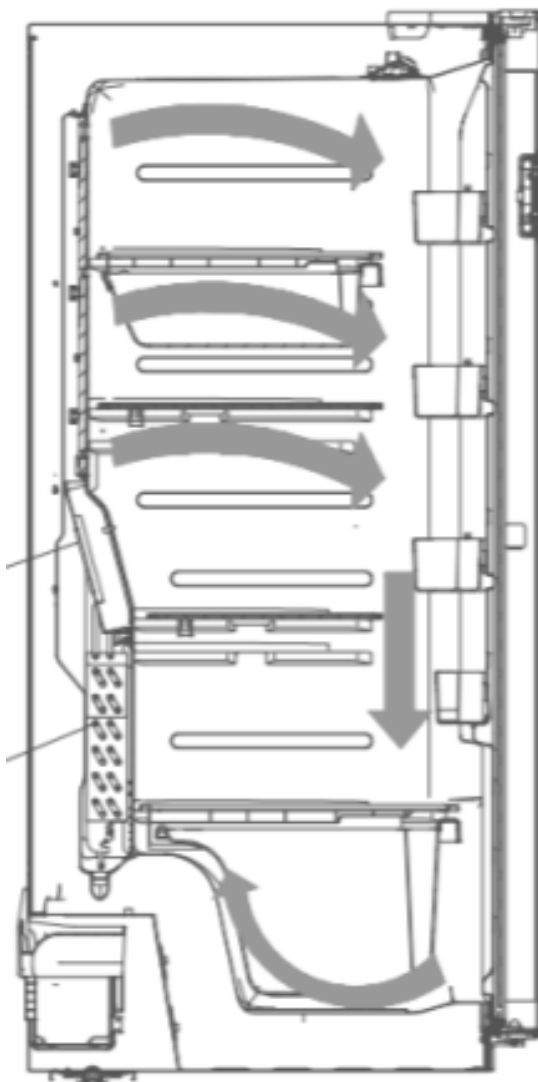
① Compressor→② Right condenser→③ Anti-condensation tube→④ Left Condenser→⑤ Dry filter→⑥ Capillary tube→⑦ Evaporator→⑧ Suction tube→① Compressor



6.2 Cooling pipeline and drain pipe inside the cabinet

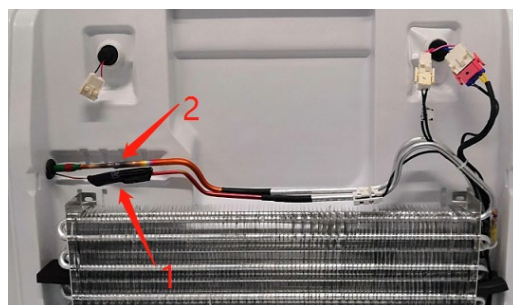


6.3 Circulating route of cooling air



6.4 Welding points in chambers or foam layer

1) Welding points on freezer evaporator



Welding point	Pipe outer diameter (mm)	
1-Freezer capillary and inlet of evaporator	Copper pipe: $\Phi 6$	Copper pipe: $\Phi 6.35$
2-Heat transition tube and outlet of evaporator	Copper pipe: $\Phi 6$	Copper pipe: $\Phi 6.35$

6.5 Welding point in the compressor case



Welding point	Pipe outer diameter (mm)	
1-Inlet of right condenser tube and outlet of venting connection tube	Steel pipe: Φ4.0	Steel pipe: Φ4.0
2-Inlet of anti-condensation tube and outlet of right condenser tube	Steel pipe: Φ4.0	Steel pipe: Φ4.0
3-Outlet of anti-condensation tube and inlet of left condenser tube	Steel pipe: Φ4. 0	Copper pipe: Φ4.0
4-Outlet of left condenser tube and inlet of dry filter	Steel pipe: Φ4. 0	Copper pipe: Φ5.0
5-Outlet of dry filter and inlet of freezer capillary	Copper pipe: Φ2.8	Copper pipe: Φ1.8
6-Heat transition tube and Suction connection tube	Copper pipe: Φ6.0	Copper pipe: Φ6.0
7-Suction connection pipe and compressor intake tube	Copper pipe: Φ6.0	Copper pipe: Φ8.17
8-Compressor outlet tube and inlet of venting connection tube	Copper pipe: Φ6.17	Steel pipe: Φ4.0
9-Compressor process tube and refrigerant filling tube	Copper pipe: Φ8.17	Copper pipe: Φ6.0

7. Dismantling of parts

7.1 Parts on the door

Door seal

Door seal is installed into door liner groove.

- 1) Open the refrigerator door;
- 2) Take the door seal out of door liner;



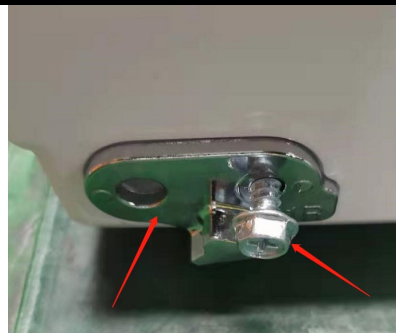
Door tray

While squeezing it inward, lift up the door tray and take it out from door liner.



Door stopper

After removing the door, remove the fixing screws at the position shown in the figure, remove the door sleeve pipe and stopper, and complete the stopper replacement.



7.2 Parts inside the refrigerator

Shelves

Lift up the division plate with a proper force and pull it out towards yourself;



Freezer Drawer

The drawer is located at the freezing chambers;

- 1) Pull the drawer out completely;
- 2) Lift it up slightly and take it out from the refrigerator.

**7.3 Light system****Light**

Light of the freezing

- 1) Remove the lamp cover



- 2) Disconnect the terminal block, remove the LED
- 3) The reverse process can complete installation.

**Light switch**

- 1) Remove the screws on the hinge cover
- 2) Remove the terminal, press the light switch elastic plastic block to remove the switch

**7.4 Air duct components in refrigerating chamber and fan motor (None)****Air duct components in freezing chamber**

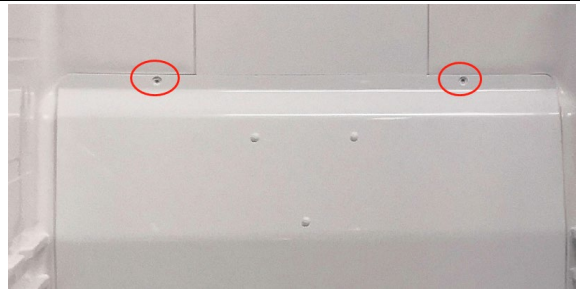
None

7.5 Air duct components in freezing chamber and fan motor

Air duct components in freezing chamber

All accessories in the freezing chamber should be dismantled before removing the air duct components.

1) Remove 1 screws on the cover plate of the freezing air duct using a cross screwdriver



2) Open the cover plate, pull out the terminal and take out the lower air duct cover plate.



3) hold the upper cover from the bottom, pull out the upper cover



4) Take out the freezing air duct assembly.



Fan motor of air duct

1) Pry up all plastic hooks, and separate the back cover plate from air duct assembly.



2) Use cross screwdriver to remove the 3 pcs screws..

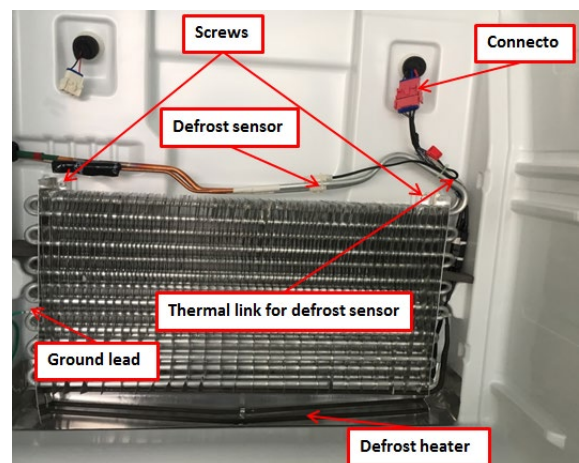


3) Take off the fan motor, replace a new one, the reverse operation is for assembly.

7.6 Evaporator and Defrost system

Evaporator in freezing chamber

- 1) Remove the air duct components in freezing and variable temperature chamber.
- 2) Disconnect all connecting terminals.
- 3) Remove the welding on inlet and outlet tubes.
- 4) Remove the evaporator.

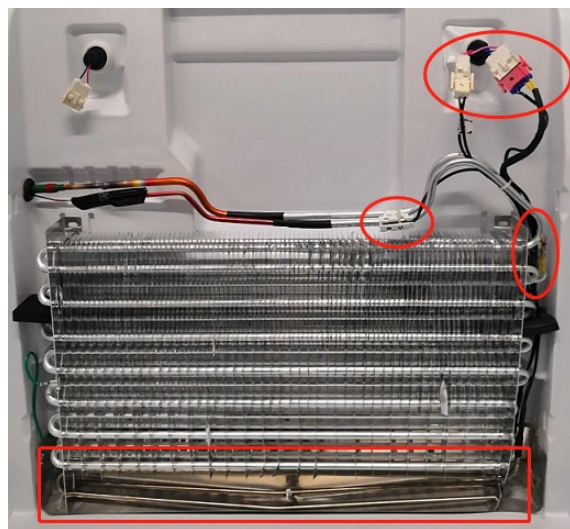


Components on the evaporator

Defrost heater with defrost sensor and fuse, it can be replaced separately.

- 1) Cut off wiring terminal
- 2) Cut off the band which fixes the sensor

- 3) Cut off the band which fixes the fuse
- 4) Separate the sensor and the evaporator
- 5) Take off the defrost heater from the supporting plate of evaporator



Evaporator in refrigerating chamber

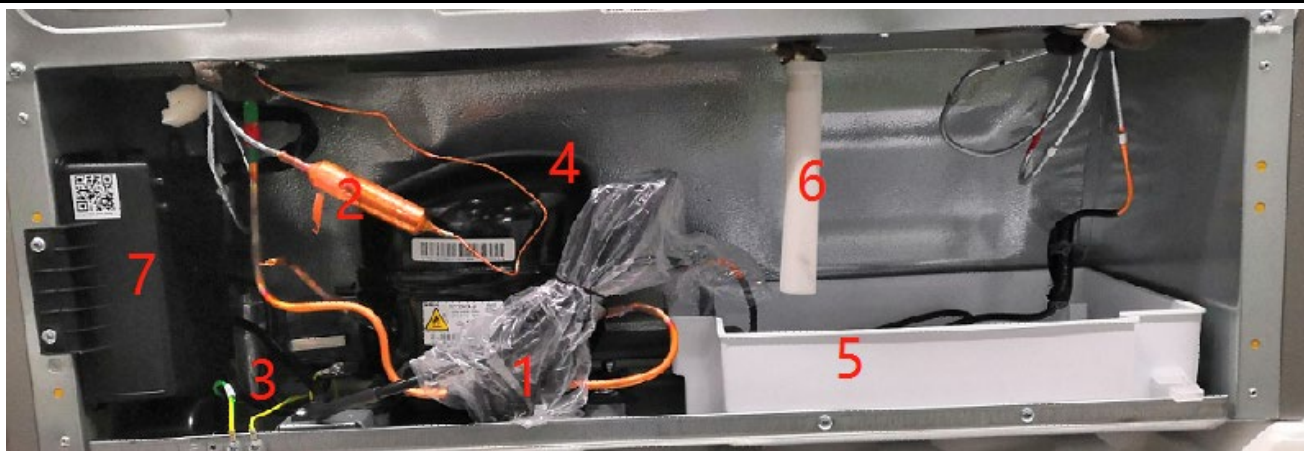
None

7.7 Compressor case

Rear cover

None

Piping and parts in the compressor case

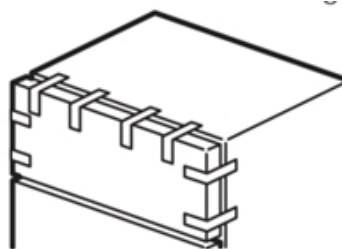


- 1-Power cord
- 2-Dry filter
- 3-Capacitor
- 4-Compressor

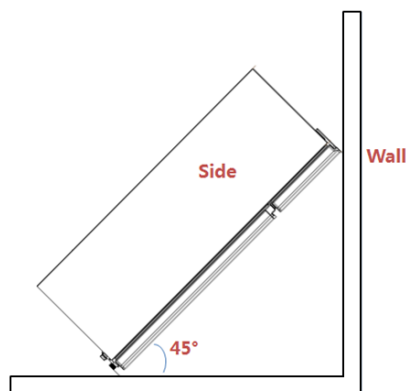
- 5-Drain tray
- 6-Drain-pipe
- 7-Main PCB mounting box

Disassembly and assembly of compressor

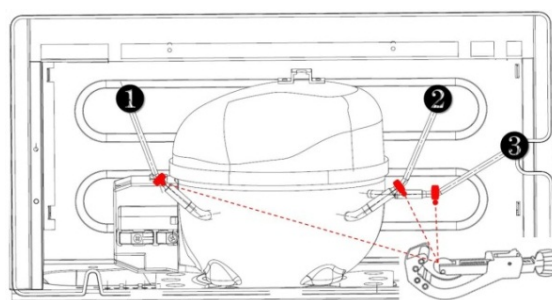
- 1) Cut off the power, remove the goods in the refrigerator, with the tape to make the door fixed firmly and prevent the door dropping when the refrigerator dumping.



2) Slowly tilt the refrigerator forward, relying on the wall or a solid enough object, leaving space to facilitate the operation. For safety, it should be carried by someone to prevent its falling.

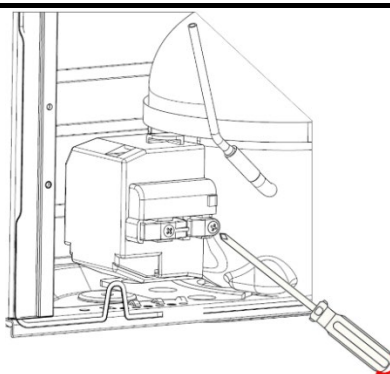


3) Cut off the compressor pipeline.-①Cut off the process pipeline.-②Cut off the low-pressure muffler.-③Cut off the high-pressure exhaust pipe.



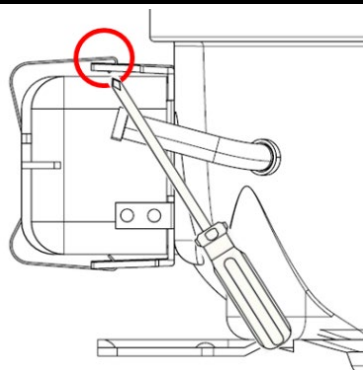
4-1) Remove the screws(for some models)

- Two screws outside
- One screw inside



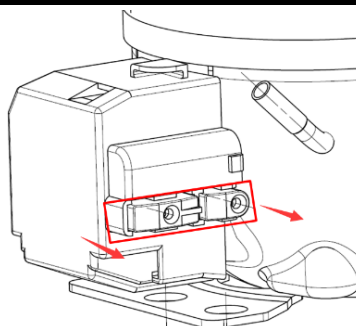
4-2) Remove the metal clamp(for some models)

- Disassembly the metal clamp that is fix the electric appliance shield



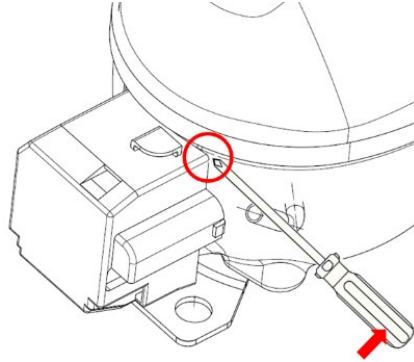
5) Remove the clipping strip

Slowly pull it out

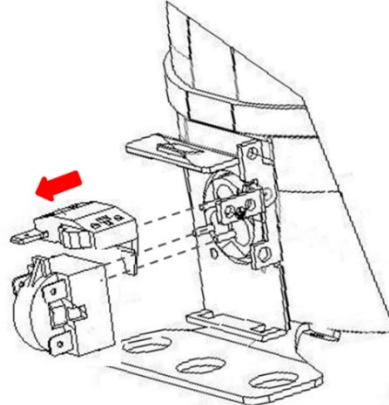


6) Remove the protective cover

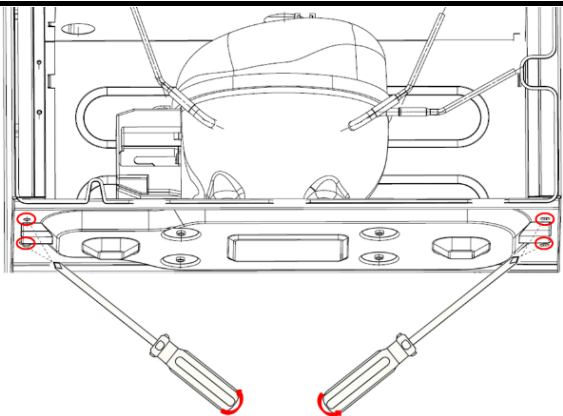
- Pry the protective cover slowly from the upper part,
- Pull it out and remove it.

**7) Remove the starter and protector**

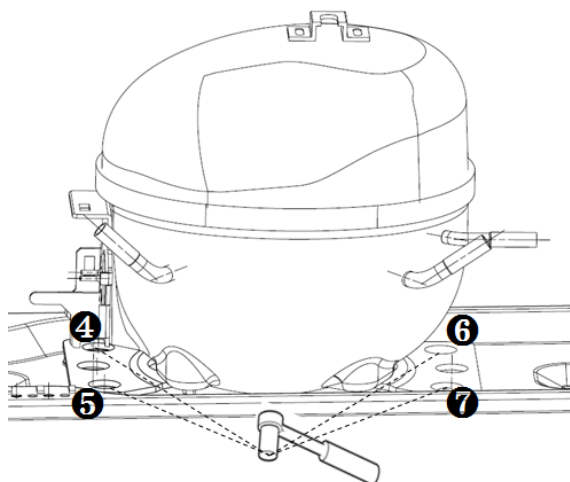
- Unplug the starter and protector (you can use a screwdriver to pry it slowly)



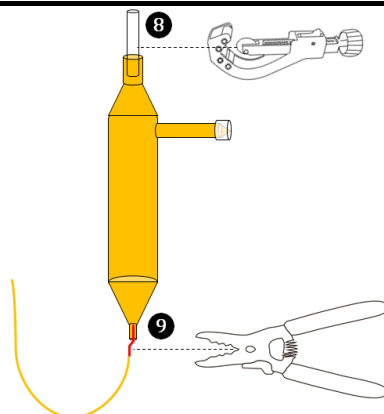
- 8) Loosen the screw of the compressor bottom plate, remove the floor together with the compressor from the box.



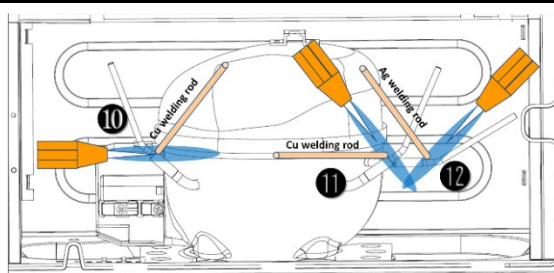
- 9) Use the wrench to remove the bolts by steps ④ ⑤ ⑥ ⑦, replace the compressor and reverse process can complete installation.



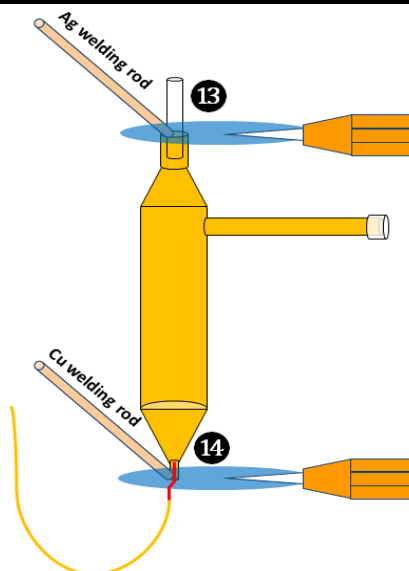
10) Use Pipe cutter cut off the condenser tube ⑧, then Shear off capillary ⑨ by the capillary tube scissors.



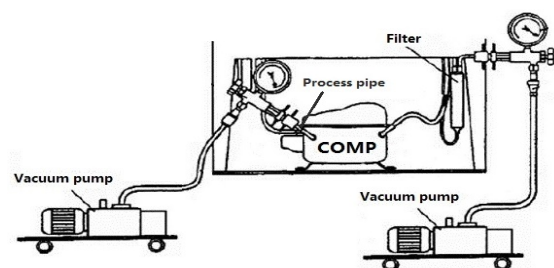
11) Replace the compressor and welding the compressor pipeline.-⑩Welding the process pipeline.-⑪Welding the low-pressure muffler.-⑫Welding the high-pressure exhaust pipe.



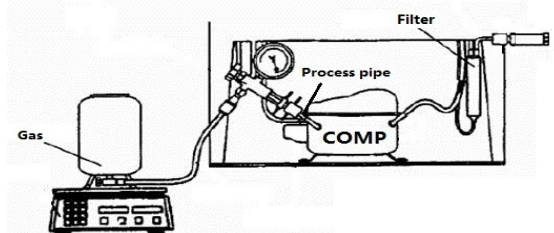
12) Replace the filter, Cu-Fe tubes welding ⑬ used Ag welding rod, Cu-Cu tubes welding ⑭ used Cu welding rod.



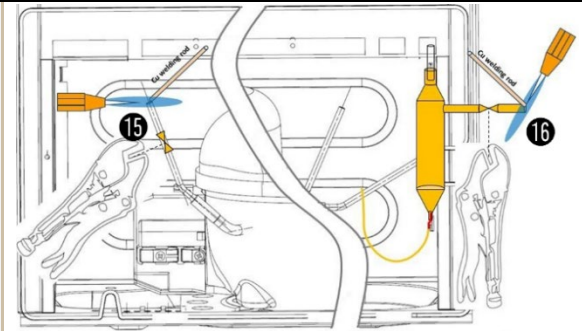
13) Vacuum system, The degree of vacuum below 6Pa.



14) Perfusion refrigerant.



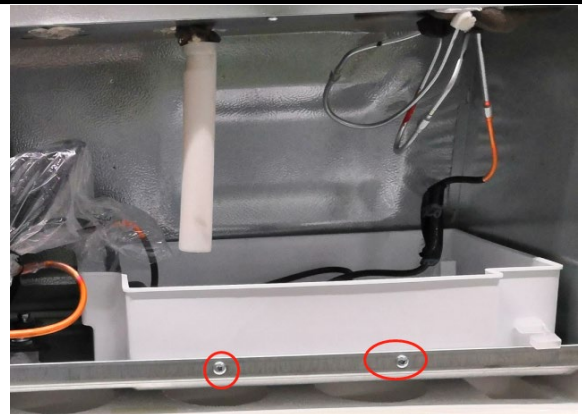
15) Use the vise grip pliers clamp the middle of the process pipe, then seal welding process tube 15 16.



Back-hanging wire tube condenser (None)

Drain tray

Remove the two fixing screws on the drain tray and pull it out, replace a new one.



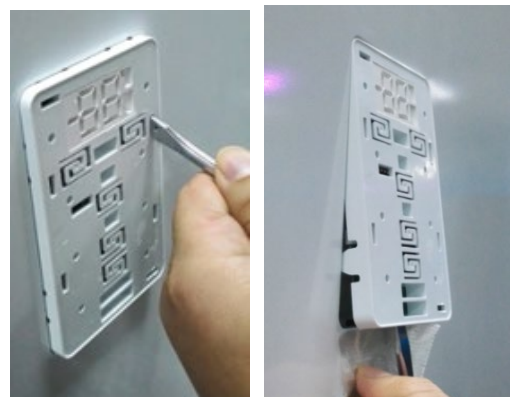
7.8 Display control board

Display control board

1) Remove the label stuck on the display control panel



2) Use a slotted screwdriver to pry open the three clips

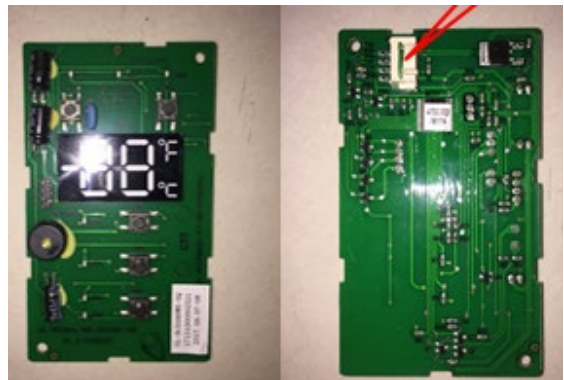


3) Remove 2 screws



4) Pull out the connector terminal and replace the display control board.

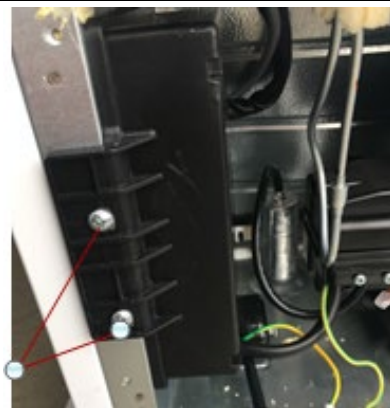
*The temperature control indication sticker is likely to be damaged when replacing, thus it is advised to prepare one for standby before replacement



7.9 Main control board

Main control board

1) Remove 2 fitting screws of main control panel mounting box

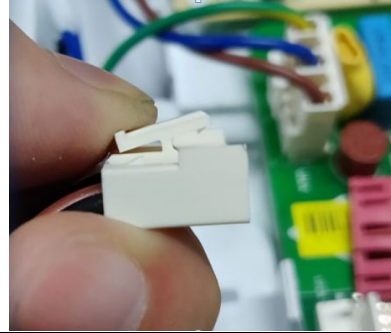
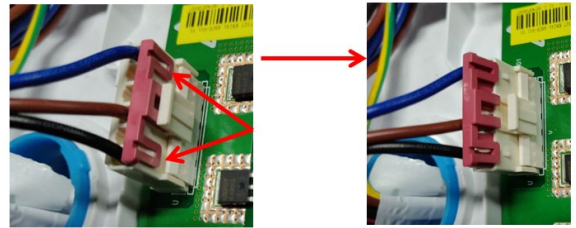


2) Take out the mounting box, pry up the cover, disconnect the wiring terminal.

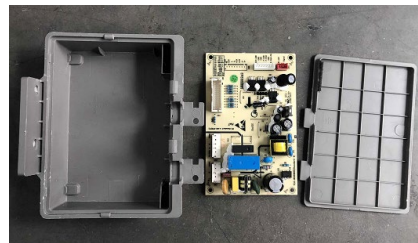


The connecting terminals remove:

- a. Use a needle tool to remove the lock at the arrow and remove the lock upward;
- b. After the lock is removed, press the hook to remove the wiring connectors



3) Remove the screws, take out the main control board, and complete the replacement.



7.10 Variable frequency driver board (None)

7.11 Water dispenser (None)

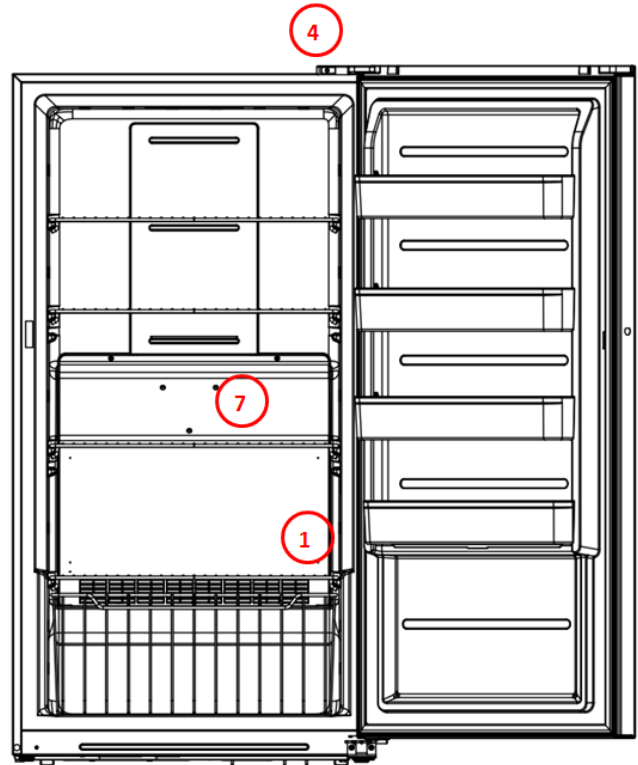
8. Temperature sensing system

8.1 Position of sensors

Position of sensors

Have 3 sensors

- ① Sensor in freezing chamber
- ② Sensor in refrigerating chamber
- ③ Sensor in Variable temperature chamber
- ④ Ambient temperature or humidity sensor
- ⑤ Ice machine sensor
- ⑥ Defrost sensor in refrigerating chamber
- ⑦ Defrost sensor freezing chamber



8.2 Replacement of sensors

Sensor in freezer chamber

The sensor is located at right of freezer chamber

- 1) Pull out the cover

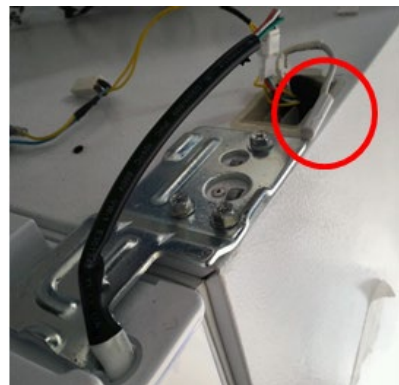


2) Pull out the sensor



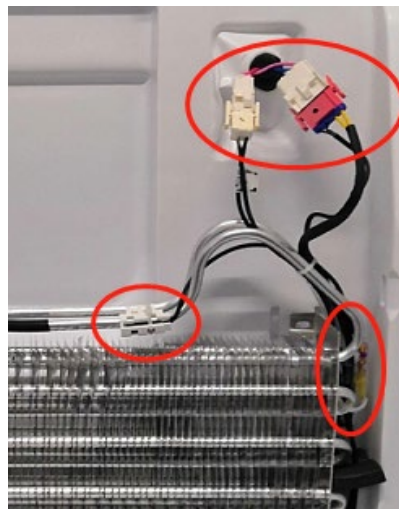
Ambient temperature sensor

The sensor is located in the wire box of the refrigerator top.

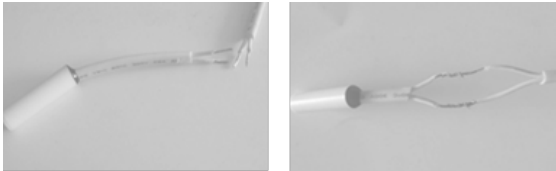
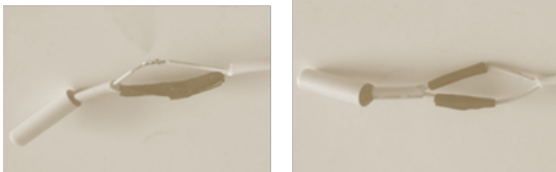


Defrost sensor in freezing chamber

- 1) Cut off wiring terminal;
- 2) Cut off the band which fixes the sensor;



8.3 Sensor without terminal replacement

Sensor replacement guidelines	
Cut off the damaged head of sensor.	
Strip off the sensor wiring.	
Take out a new sensor to cut the head of sensor. (Spare parts code: 11201007000795) Its technical specifications apply to all MIDEA refrigerators.	
Strip off the head of the sensor and connect it.	
Wrap the two wires together with insulation tape.	
Wrap the two wires together.	

8.4 Sensor R/T table

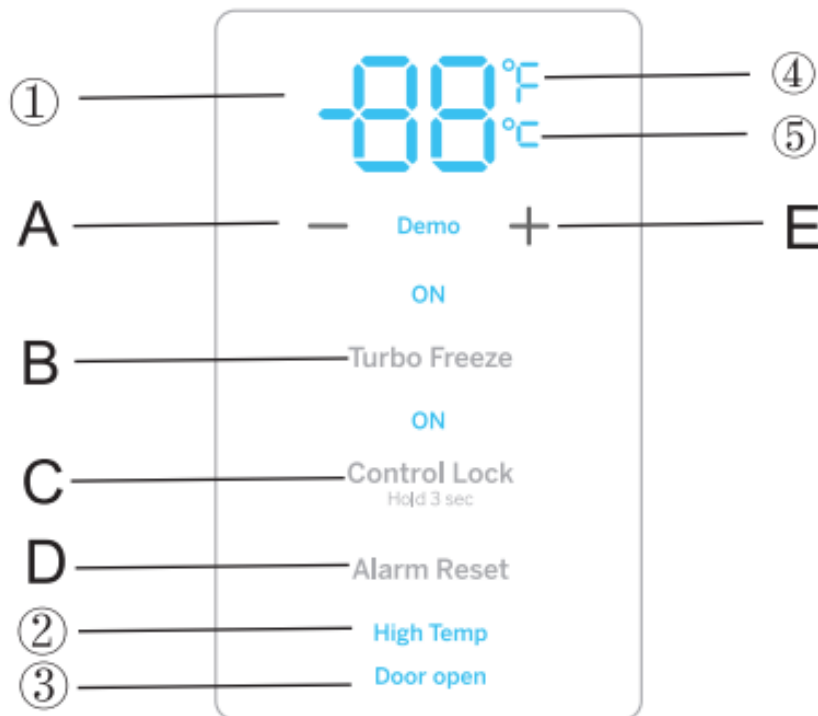
T _x (°C)	T _x (°F)	R (KΩ)	T _x (°C)	T _x (°F)	R (KΩ)	T _x (°C)	T _x (°F)	R (KΩ)
-30	-22.00	33.81	-5	23.00	8.392	20	68.00	2.501
-29	-20.20	31.85	-4	24.80	7.968	21	69.80	2.391
-28	-18.40	30.01	-3	26.60	7.568	22	71.60	2.287
-27	-16.60	28.29	-2	28.40	7.190	23	73.40	2.188
-26	-14.80	26.68	-1	30.20	6.833	24	75.20	2.094
-25	-13.00	25.17	0	32.00	6.495	25	77.00	2.005
-24	-11.20	23.76	1	33.80	6.175	26	78.80	1.919
-23	-9.40	22.43	2	35.60	5.873	27	80.60	1.838
-22	-7.60	21.18	3	37.40	5.587	28	82.40	1.761
-21	-5.80	20.01	4	39.20	5.315	29	84.20	1.687
-20	-4.00	18.90	5	41.00	5.060	30	86.00	1.617
-19	-2.20	17.87	6	42.80	4.818	31	87.80	1.550
-18	-0.40	16.90	7	44.60	4.589	32	89.60	1.486
-17	1.40	15.98	8	46.40	4.372	33	91.40	1.426
-16	3.20	15.12	9	48.20	4.167	34	93.20	1.368
-15	5.00	14.310	10	50.00	3.972	35	95.00	1.312
-14	6.80	13.550	11	51.80	3.788	36	96.80	1.259
-13	8.60	12.830	12	53.60	3.613	37	98.60	1.209
-12	10.40	12.160	13	55.40	3.447	38	100.40	1.161
-11	12.20	11.520	14	57.20	3.290	39	102.20	1.115
-10	14.00	10.920	15	59.00	3.141	40	104.00	1.071
-9	15.80	10.350	16	60.80	2.999	41	105.80	1.029
-8	17.60	9.820	17	62.60	2.865	42	107.60	0.989
-7	19.40	9.316	18	64.40	2.737	43	109.40	0.951
-6	21.20	8.841	19	66.20	2.616	44	111.20	0.914

9. Function and operation

9.1 Display operation panel

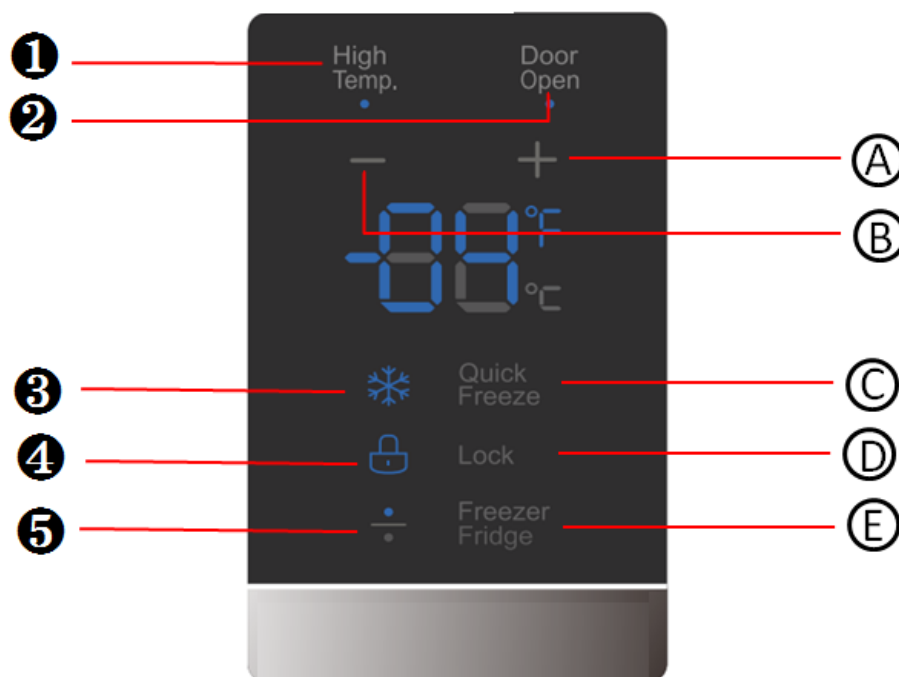
Single freezing display screen

Button	Icons
A. Temperature down	① Temperature display area
B. Quick freezer	② High temperature alarm
C. Lock and unlock	③ Door opening alarm
D. Release alarm	④ Fahrenheit temperature
E. Temperature up	⑤ Celsius temperature



Refrigeration / freezing conversion display screen

Icons	Button
① High temperature alarm	Ⓐ Temperature up
② Door opening alarm	Ⓑ Temperature down
③ Quick freezing	Ⓒ Quick freeze
④ Locking	Ⓓ Locking and unlock
⑤ Refrigerating and freezing switching	Ⓔ Refrigerating and freezing change



9.2 Temperature control

A. Single freezing model

Temperature setting of freezing chamber

The temperature setting range of freezer chamber is $-24^{\circ}\text{C} \sim -16^{\circ}\text{C}$ ($10^{\circ}\text{F} \sim -11^{\circ}\text{F}$). The temperature can be set circularly, press the button “-” or “+” once, the temperature turn by $1^{\circ}\text{C}/^{\circ}\text{F}$. The setting is effective after being locked.

$-14^{\circ}\text{C} \rightarrow -13^{\circ}\text{C} \rightarrow \dots \rightarrow -20^{\circ}\text{C} \rightarrow -21^{\circ}\text{C} \rightarrow -20^{\circ}\text{C} \rightarrow \dots \rightarrow -14^{\circ}\text{C}$

$6^{\circ}\text{F} \rightarrow 5^{\circ}\text{F} \rightarrow \dots \rightarrow -5^{\circ}\text{F} \rightarrow -6^{\circ}\text{F} \rightarrow -5^{\circ}\text{F} \rightarrow \dots \rightarrow 5^{\circ}\text{F} \rightarrow 6^{\circ}\text{F}$

B. Refrigeration / freezing conversion model

9.2.1 Temperature setting of refrigerating chamber

The temperature setting range of refrigerating chamber is $2^{\circ}\text{C} \sim 8^{\circ}\text{C}$ ($35^{\circ}\text{F} \sim 46^{\circ}\text{F}$), the temperature can be set circularly, press the button “V” or “^” once, the temperature turn by $1^{\circ}\text{C}/^{\circ}\text{F}$. The setting is effective after being locked.

$8^{\circ}\text{C} \rightarrow 7^{\circ}\text{C} \rightarrow \dots \rightarrow 2^{\circ}\text{C} \rightarrow 3^{\circ}\text{C} \rightarrow \dots \rightarrow 8^{\circ}\text{C}$

$46^{\circ}\text{F} \rightarrow 45^{\circ}\text{F} \rightarrow 44^{\circ}\text{F} \rightarrow \dots \rightarrow 35^{\circ}\text{F} \rightarrow 36^{\circ}\text{F} \rightarrow \dots \rightarrow 46^{\circ}\text{F}$

9.2.2 Temperature setting of freezing chamber

The temperature setting range of freezer chamber is $-24^{\circ}\text{C} \sim -16^{\circ}\text{C}$ ($10^{\circ}\text{F} \sim -11^{\circ}\text{F}$). The temperature can be set circularly, press the button “V” or “^” once, the temperature turn by $1^{\circ}\text{C}/^{\circ}\text{F}$. The setting is effective after being locked.

$-12^{\circ}\text{C} \rightarrow -13^{\circ}\text{C} \rightarrow \dots \rightarrow -24^{\circ}\text{C} \rightarrow -23^{\circ}\text{C} \rightarrow -22^{\circ}\text{C} \rightarrow \dots \rightarrow -12^{\circ}\text{C}$

$10^{\circ}\text{F} \rightarrow 9^{\circ}\text{F} \rightarrow 8^{\circ}\text{F} \rightarrow \dots \rightarrow -10^{\circ}\text{F} \rightarrow -11^{\circ}\text{F} \rightarrow -10^{\circ}\text{F} \rightarrow \dots \rightarrow 9^{\circ}\text{F} \rightarrow 10^{\circ}\text{F}$

9.3 Mode setting

9.3.1 Super freeze mode

Press the “**Turbo Freeze**” button for three seconds and the “ON” icon will illuminate. The temperature of freezing chamber is set to -24°C (-11°F). Automatically exit for up to 8 hours.

9.3.2 Celsius / Fahrenheit temperature switch

Press the “+” and “-” button for three seconds to switch, and the $^{\circ}\text{C}$ and $^{\circ}\text{F}$ icon will illuminate.

9.3.3 Locking and unlocking

In unlocking state, keep pressing the “**Unlock**” button for 3 seconds to lock, and the locking beeping will be rang.

In locking state, keep pressing the “**Unlock**” button for 3 seconds to unlock, and the unlocking beeping will be rang.

The refrigerator enters into locking state automatically 30 seconds after no button operation.

Except for the door open alarm and failure alarm can be carried out in the locking state, others button operations shall be conducted in unlocking state.

9.3.4 High temperature alarm

When $T_f \geq -8^{\circ}\text{C}$ and duration ≥ 36 hours. ($< -8^{\circ}\text{C}$ and the duration is cleared) (It must be detected once that the temperature of frozen sensor is lower than -15°C for judgement), then enter the over-temperature alarm, the high-temperature alarm indicator lights up; the buzzer sounds for 10 seconds at 1Hz frequency and then sounds for 10 seconds every 30 minutes.

The alarm is released when $T_f < -12^{\circ}\text{C}$. Operate any key to cancel the buzzer alarm.

9.4 Defrosting function

◆ Defrosting theory:

The defrosting of evaporator is realized by the heating of heater, following the temperature rise, the frost on evaporator becomes water, and the water flow into the evaporating pan via the draining system, the water in evaporating pan evaporate away finally

◆ Defrosting steps:

Compressor shutdown--- the freezer fan motor closed, the heater start working ---the heater stop when the temperature rise to setting--- after 7 minutes, the compressor will start--- after 8 minutes, the fan motor will start work

◆ Meet one of the following conditions , defrost heating exit:

- 1) When the defrosting sensor has no fault, the measured temperature $T_{fd} \geq$ the set temperature 12°C (54°F) which frozen defrost heating exit and the set temperature 7°C (45°F) which refrigerat defrost heating exit
- 2) When the defrosting sensor has no fault, the defrosting time is ≥ 60 minutes
- 3) When the defrosting sensor is faulty , the heater will stop after work 20 minutes

9.5 Open door alarm

When the door is opened, the door opening indicator light is on, and when the door is closed, the door opening indicator light goes out;

At any time, when the door opening time of the freezing chamber is more than 5 minutes, it will enter the door opening alarm, and the door opening alarm indicator light flashes according to 500ms opening and

500ms closing; The buzzer sounds with frequency of once per second or pressing any key can eliminate the buzzer alarm.

Note: The door opening alarm time can be set through the APP and then remembered to the main board after setting.

9.6 Error code and solutions

Note: Midea full common fault code, combined with the actual product display reference.

Error code	Fault Type	Troubleshooting and Solutions
E0	fault of ice maker	Step 1: Check whether the connection terminals on the ice maker and the main PCB are plugged in place and whether there are foreign matters in them; after cleaning the terminals, plug them in again. Step 2: Enter the forced ice making mode and check whether the ice maker works normally. Step 3: If the ice maker does not work, replace the ice motor. Step 4: If the fault still occurs, replace the main PCB.
E1	Temperature sensor fault in refrigerating chamber	Step 1: Check whether the connection terminals are plugged in place and whether there are foreign matters in them; after cleaning the terminals, plug them in again. Step 2: If the fault still occurs, pull out the corresponding connection terminal on the main PCB, use a multimeter to check the resistance value of the sensor, and confirm whether it is normal.
E2	Temperature sensor fault in freezing chamber	Step 3: If the resistance value is wrong, replace the sensor. Step 4: If the fault still occurs, replace the main PCB.
E3	Temperature sensor fault in variable chamber	
E4	Defrost sensor fault in refrigerating chamber	Step 1: Check whether the connection terminals are plugged in place and whether there are foreign matters in them; after cleaning the terminals, plug them in again. Step 2: If the fault still occurs, pull out the corresponding connection terminal on the main PCB, use a multimeter to check the resistance value of the sensor, and confirm whether it is normal.
E5	Defrost sensor fault in freezing chamber	Step 3: If the resistance value is wrong, replace the sensor. Step 4: If the fault still occurs, replace the main PCB.
E6	Communication failure	Step 1: Check whether the connection terminal on the display control panel, hinge cover and main PCB are plugged in place and whether there are foreign matters in them; after cleaning the terminals, plug them in again. Step 2: If the fault still occurs, pull out all connection terminals, use a multimeter to check the resistance value of the wire between the

		display control board and the main PCB to see if it is broken. If test value is $\infty \Omega$, the wire is broken.(If the wire in the door is broken, replace the door. Other conditions cannot be repaired.) Step 3: If the wire is OK, replace the display control board. Step 4: If the fault still occurs, replace the main PCB.
E7	Ambient temperature sensor fault	Step 1: Check whether the connection terminals are plugged in place and whether there are foreign matters in them; after cleaning the terminals, plug them in again. Step 2: If the fault still occurs, pull out the corresponding connection terminal on the main PCB, use a multimeter to check the resistance value of the sensor, and confirm whether it is normal.
E8	Defrost sensor fault in variable chamber	Step 3: If the resistance value is wrong, replace the sensor. Step 4: If the fault still occurs, replace the main PCB.
E9	High temperature alarm in freezing chamber	Step 1: Check whether the door is not closed, or whether there is leakage between the door gasket and the cabinet. Step 2: Check whether the door gasket is deformed and causes leakage, reshape or replaced with a new one. Step 3: Check whether the freezing frost is OK, if there is ice on the evaporator and the fan motor is frozen, replace a new defroster heater. Step 4: Check whether the light switch is damaged, replace new one. Step 5: Check whether the freezer fan stops working, plug and unplug connection terminals, replace a new fan motor. Step 6: Check if the pipeline is leaking or blocked. After maintenance, vacuum and refill the refrigerant.
EE	Circuit fault of ice maker sensor	Step 1: Check whether the connection terminals on the ice maker and the main PCB are plugged in place and whether there are foreign matters in them; after cleaning the terminals, plug them in again. Step 2: If the fault still occurs, pull out the connection terminal on ice maker, use a multimeter to check the resistance value of the sensor, and confirm whether it is normal. Step 3: If the resistance value is wrong, replace the sensor. Step 4: If the fault still occurs, replace the main PCB.
EH	Circuit fault of ambient humidity sensor	Step 1: Check whether the connection terminals in the hinge cover and the main PCB are plugged in place and whether there are foreign matters in them; after cleaning the terminals, plug them in again. Step 2: If the fault still occurs, pull out the connection terminal on the hinge cover, use a multimeter to check the voltage value of the sensor, and confirm whether it is normal. Step 3: If the voltage value is wrong, replace the sensor. Step 4: If the fault still occurs, replace the main PCB.

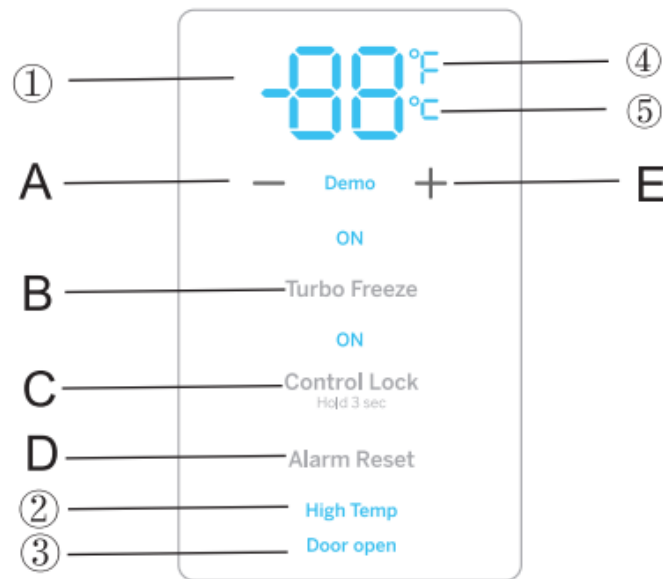
EF	Water tank installation failure	Step 1: Check whether the water tank is installed in place, pull out the water tank and install it again; Step 2: Check whether the connection terminals behind the water tank seat is plugged in place and whether there are foreign matters in it; after cleaning the terminal, plug it in again. Step 3: Check whether the switch in the water tank seat is damaged, replace a new switch.
----	---------------------------------	---

9.7 Test mode

All below functions are only for diagnosis and test purpose, we advise to restart the refrigerator by power on/off if have used these functions.

Test items	Setting Method	Setting result
Enter Test Mode	Keep pressing the “ LOCK ” and “-” button for 3 seconds and release	LED indicators display “0”, then the refrigerator enters into test mode
	After entering into test mode, if no button is pressed within 30 seconds	then the refrigerator will exit the test mode and return to normal operation mode
Select to enter into forced cooling mode	Enter into test mode and press “+” button for the first time	LED indicators display “1”, then the compressor and the fan will start working
	In forced cooling mode, if no button is pressed within 36 hours,	then the refrigerator will exit the test mode and return to normal operation mode
Select to enter into forced defrosting mode	Enter into test mode and press “ Vacation ” button for the second time	LED indicators display “3”, then the compressor and the freezing fan will stop working, the freezing defrost heater starts to work.
	In forced defrosting mode, when the freezing and flex. defrosting sensor reach a temperature of 12°C (54°F) or 7°C (45°F) and the defrosting heater has been working for at least 2 minutes.	then the refrigerator will exit the test mode and return to normal operation mode
	In forced defrosting mode, if the defrosting heater has been working for 60 minutes,	then the refrigerator will exit the test mode and return to normal operation mode
Select to exit the test mode	Enter into test mode and keep pressing the “ LOCK ” and “—” button for 3 seconds	LED indicators display “0”, then the refrigerator will exit the test mode and return to normal operation mode.

9.8 Demo mode



Press the A+B+E key at the same time for 4 seconds to enter / exit demo mode

9.9 Backup data for power fail

- 1) The running state of the refrigerator is remembered after compressor running for 1 hour continuously
- 2) The running state of the refrigerator is remembered after change function settings and lock.
- 3) When the refrigerator is out of power and recharged, the running state of the refrigerator is same as before.

10. Compressor

10.1 Compressor on and off control specifications

1.1 When one of the following conditions is met, the compressor stops:

- 1) Measured temperature in all chambers $\leq$ Shutdown temperature of all chamber
- 2) Enter defrosting cycle
- 3) The compressor runs continuously for more than 3 hours, will stop at least 10 minutes

1.2 When compressor shutdown time has been more than 5 min, one of the following conditions is met, the compressor starts up:

- 1) Measured temperature in freezer chamber $\geq$ Start up temperature of freezer
- 2) Refrigerator is set to forced cooling mode or super cooling or super freeze mode

★When 1.1 and 1.2 are not satisfied, the compressor maintains the original state

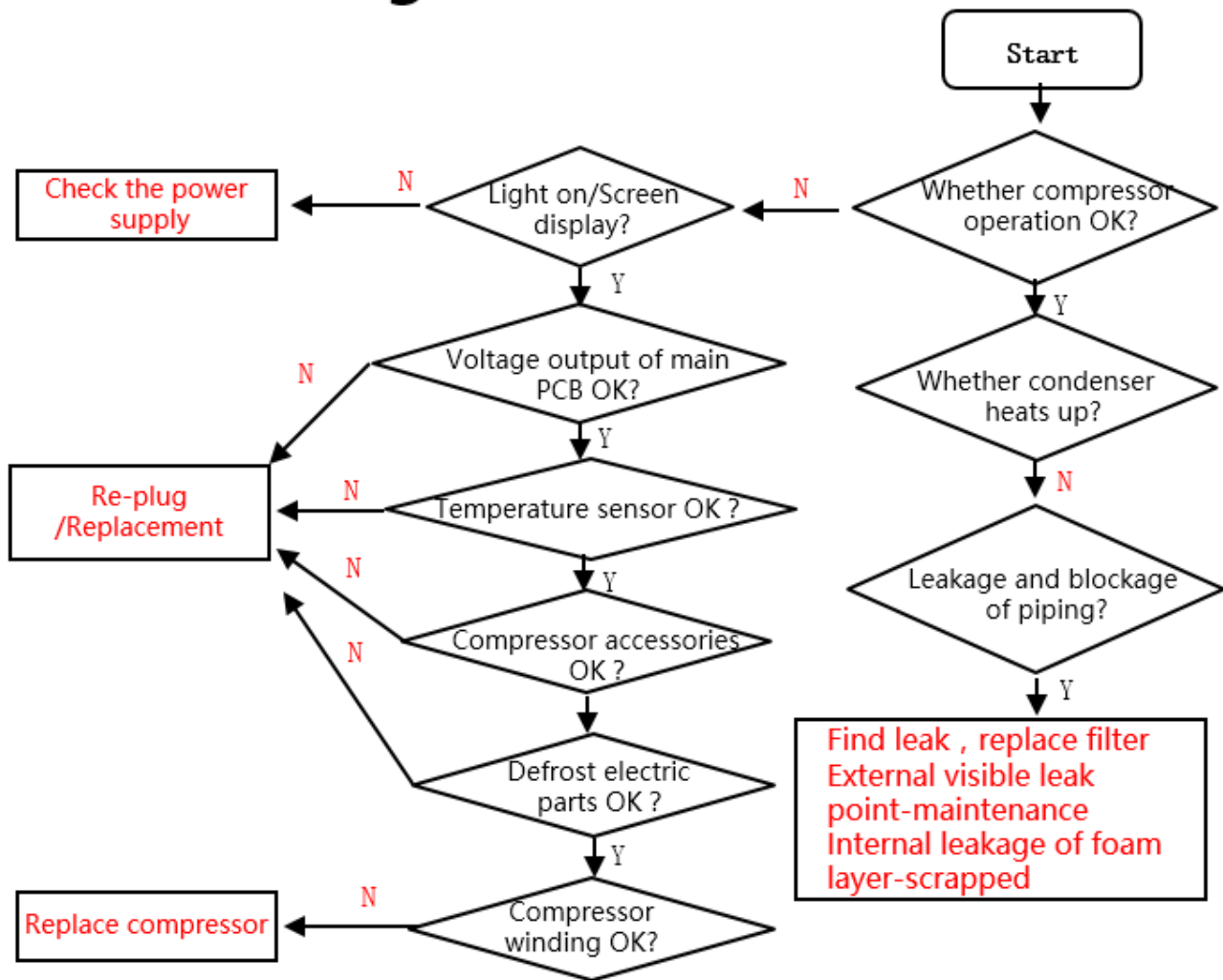
★Shutdown temperature and Start up temperature have different values due to ambient temperature.

10.2 Inverter board fault analysis (None)

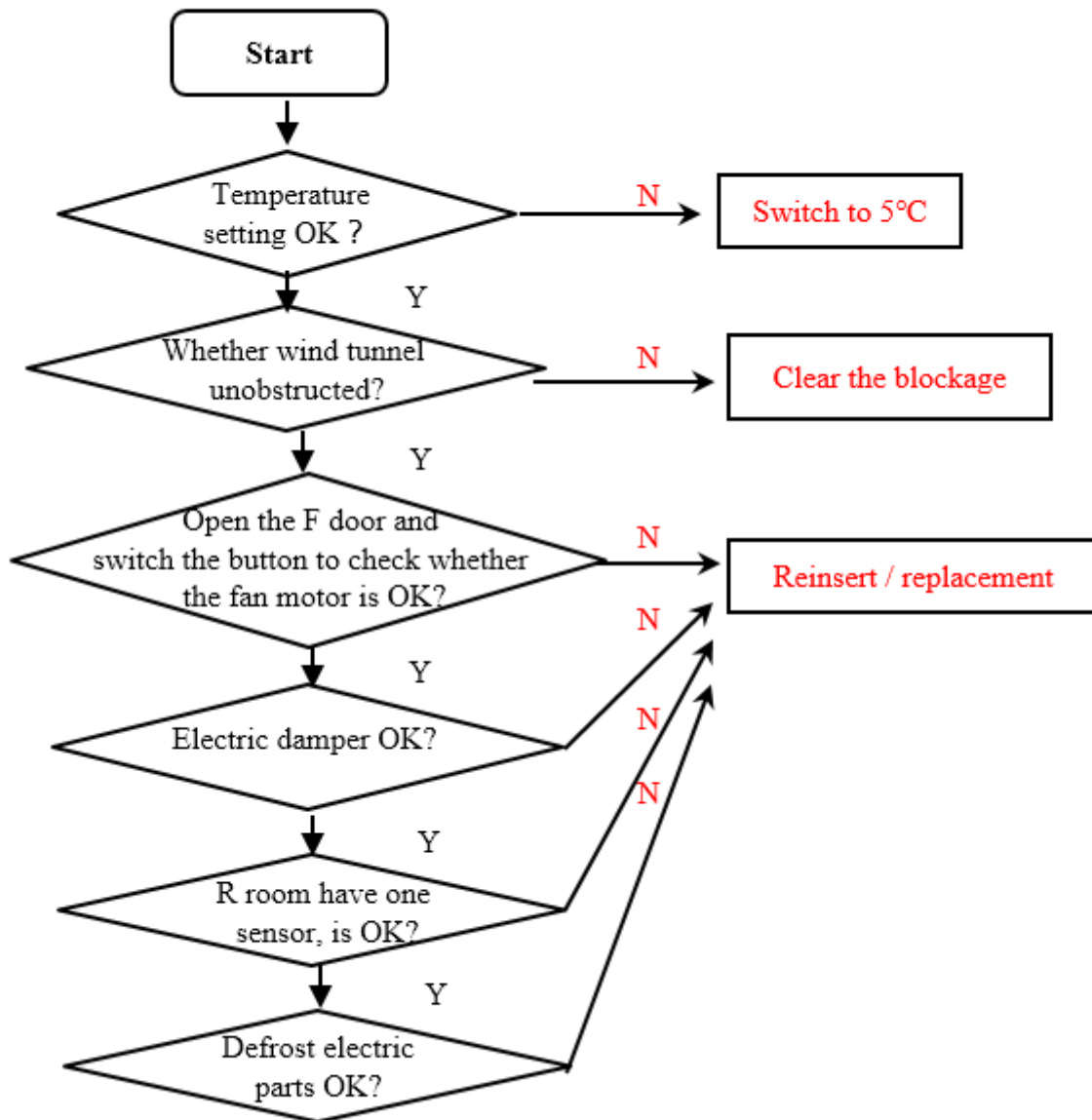
11. Troubleshooting Method

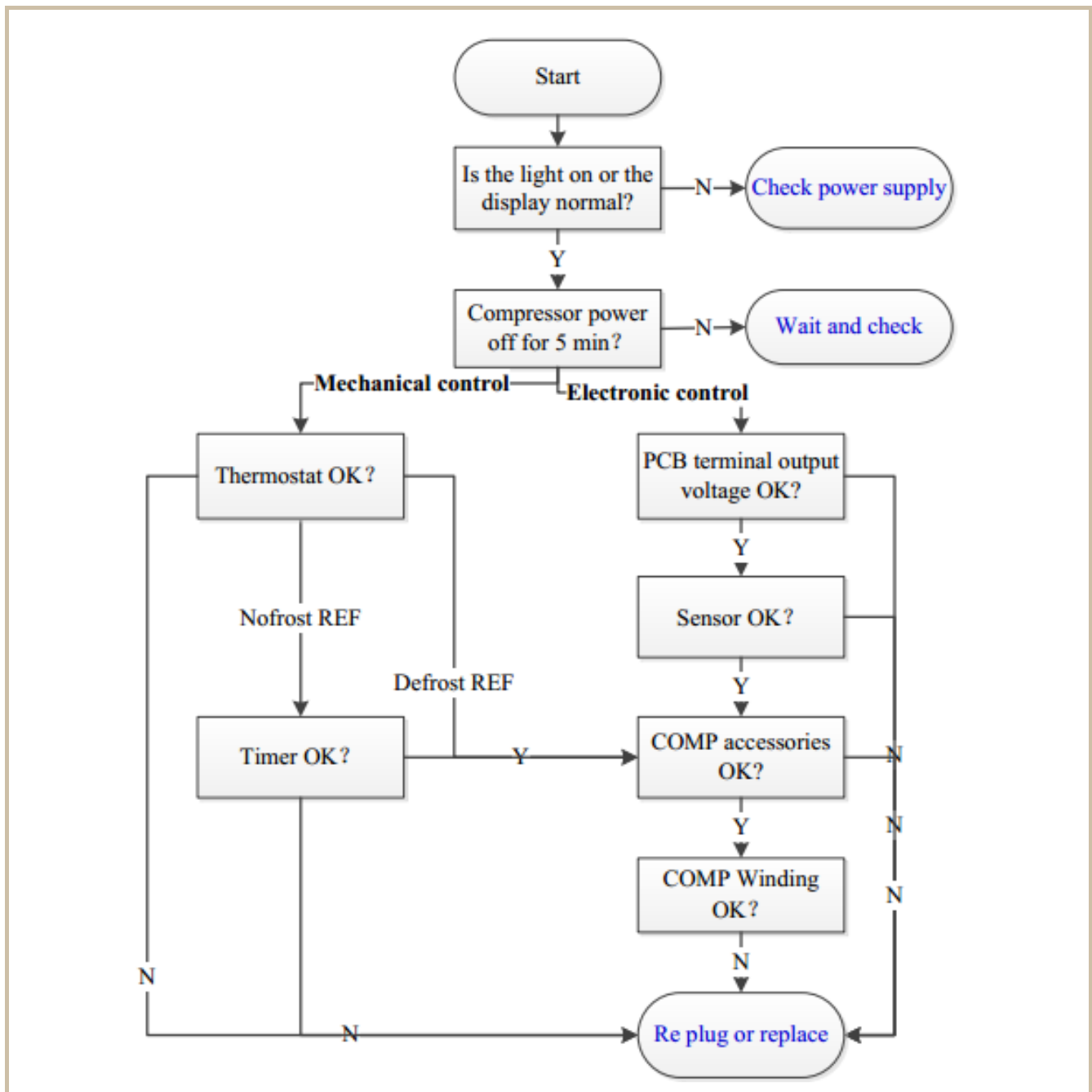
11.1 No cooling (Air cooling-Electronic)

No cooling of F room and R room



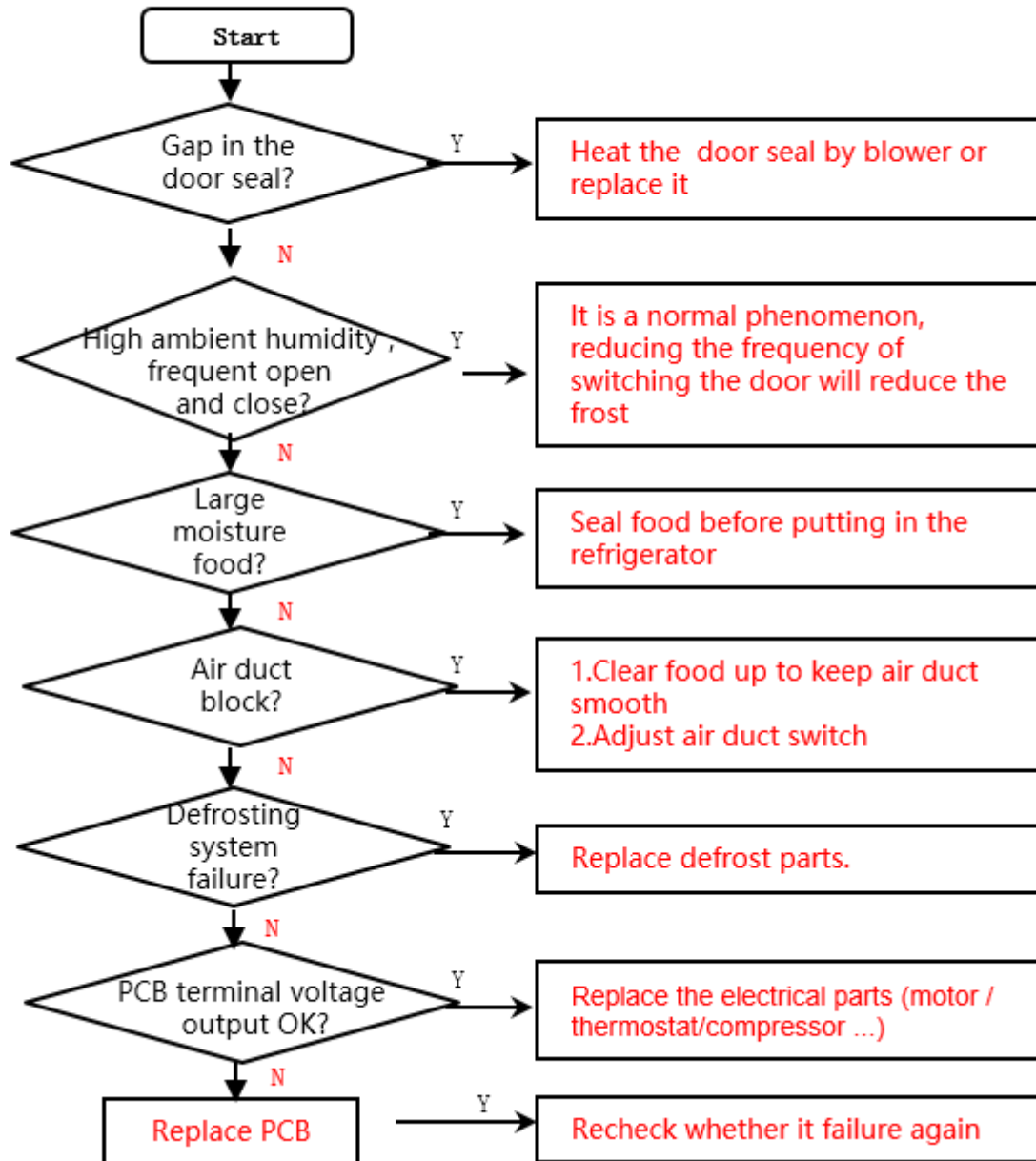
F room cooling, R room no cooling



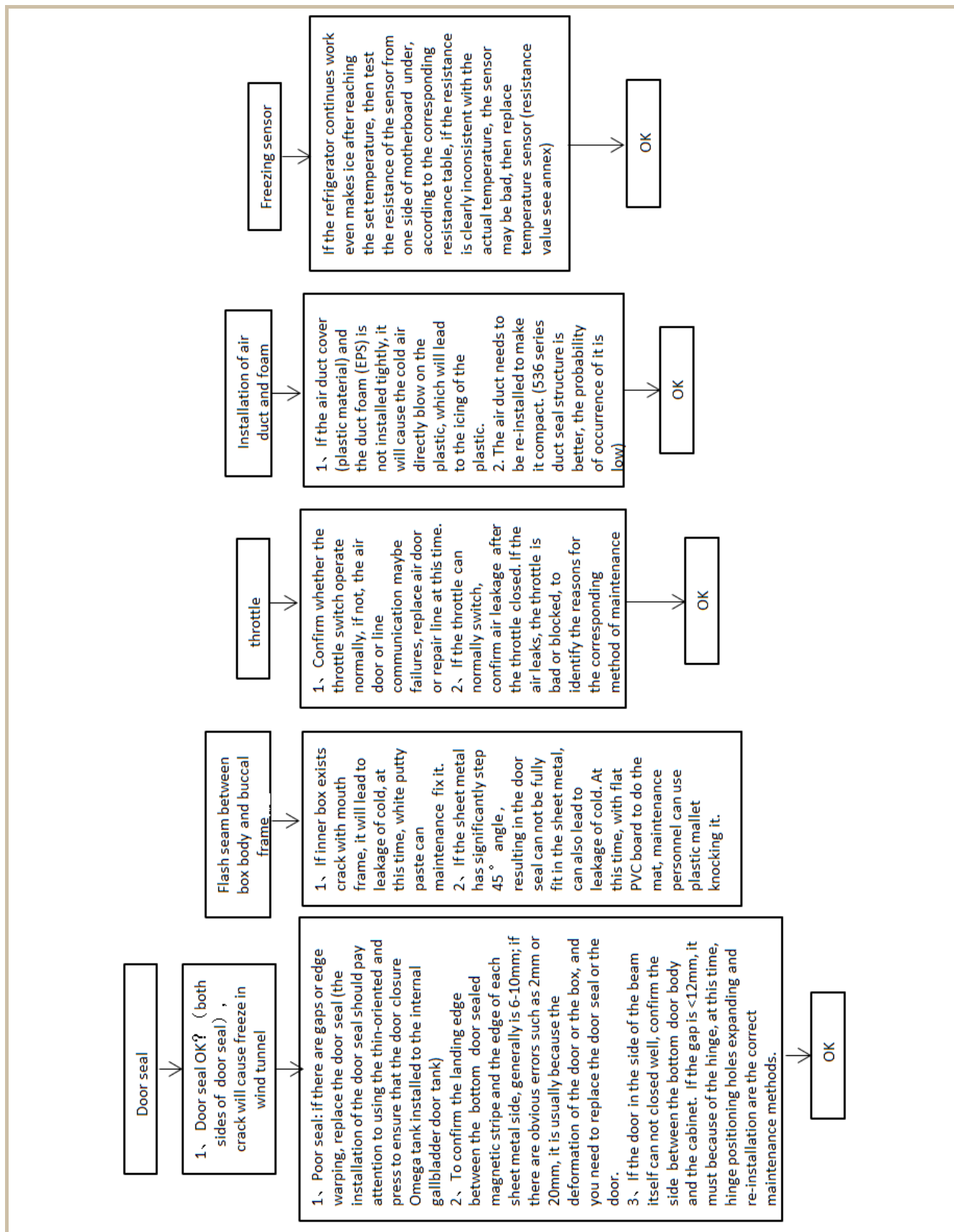
11.2 No working of compressor

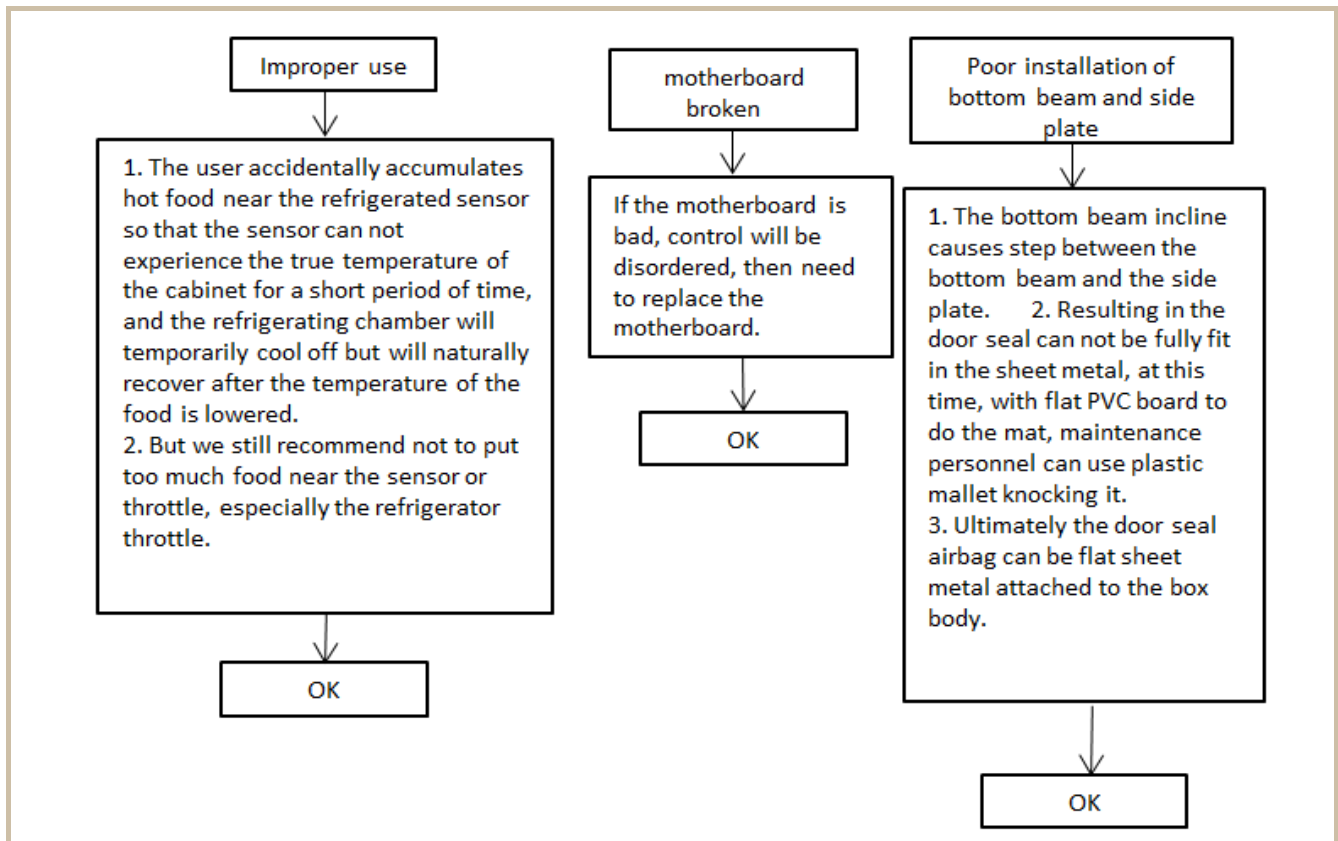
11.3 Inside frosting, no defrosting

Inside frosting, no defrosting

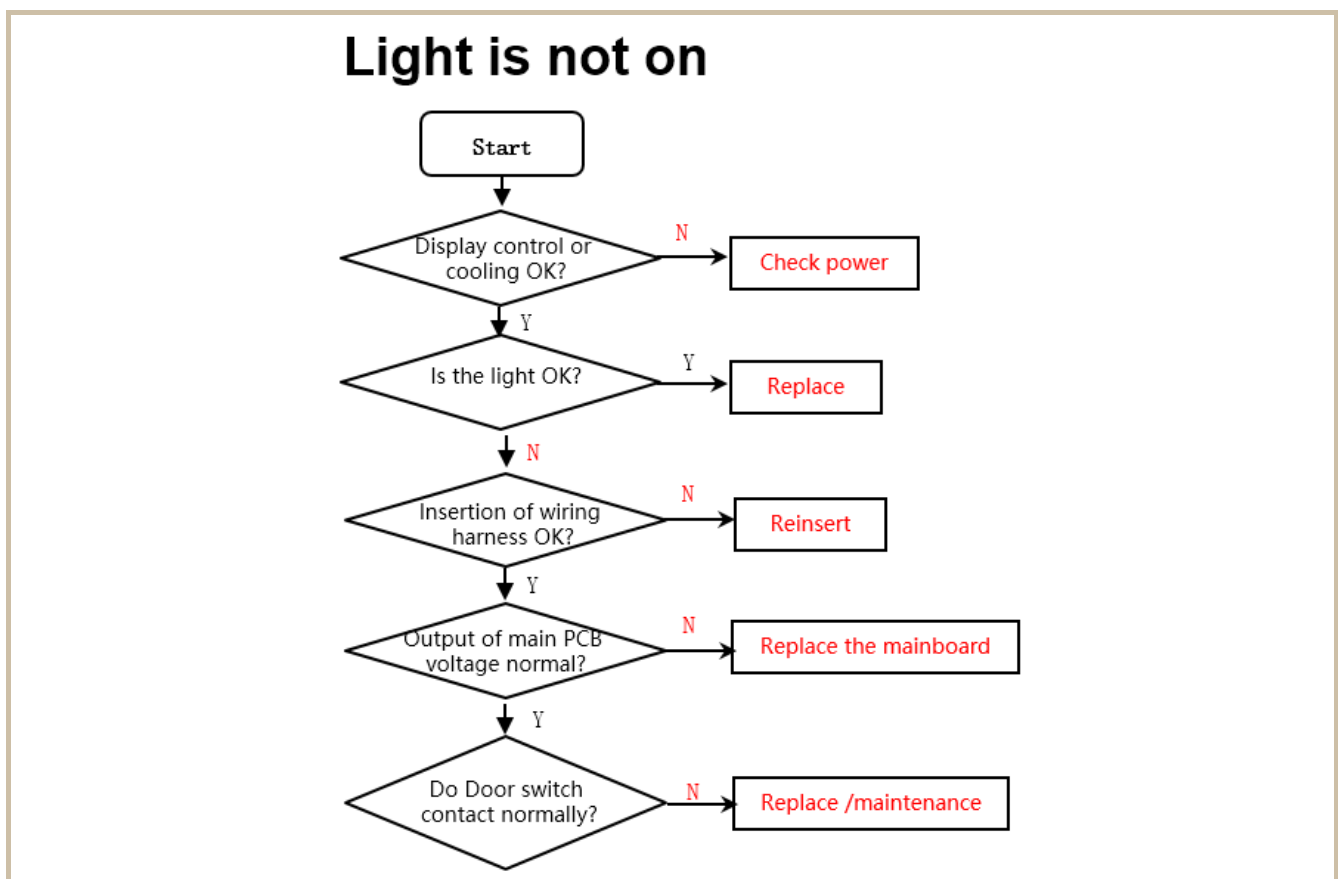


11.4 Inside frosting, no defrosting-Maintenance guidelines

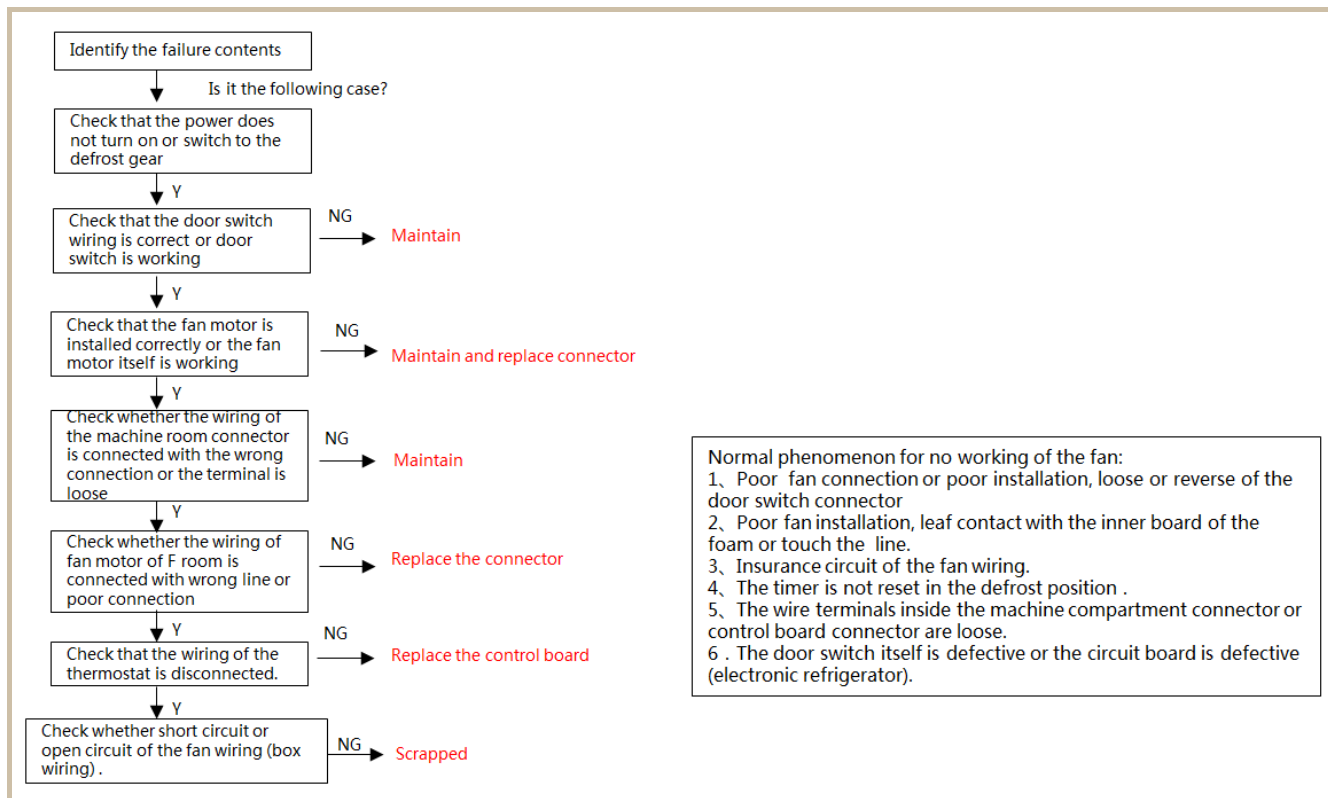




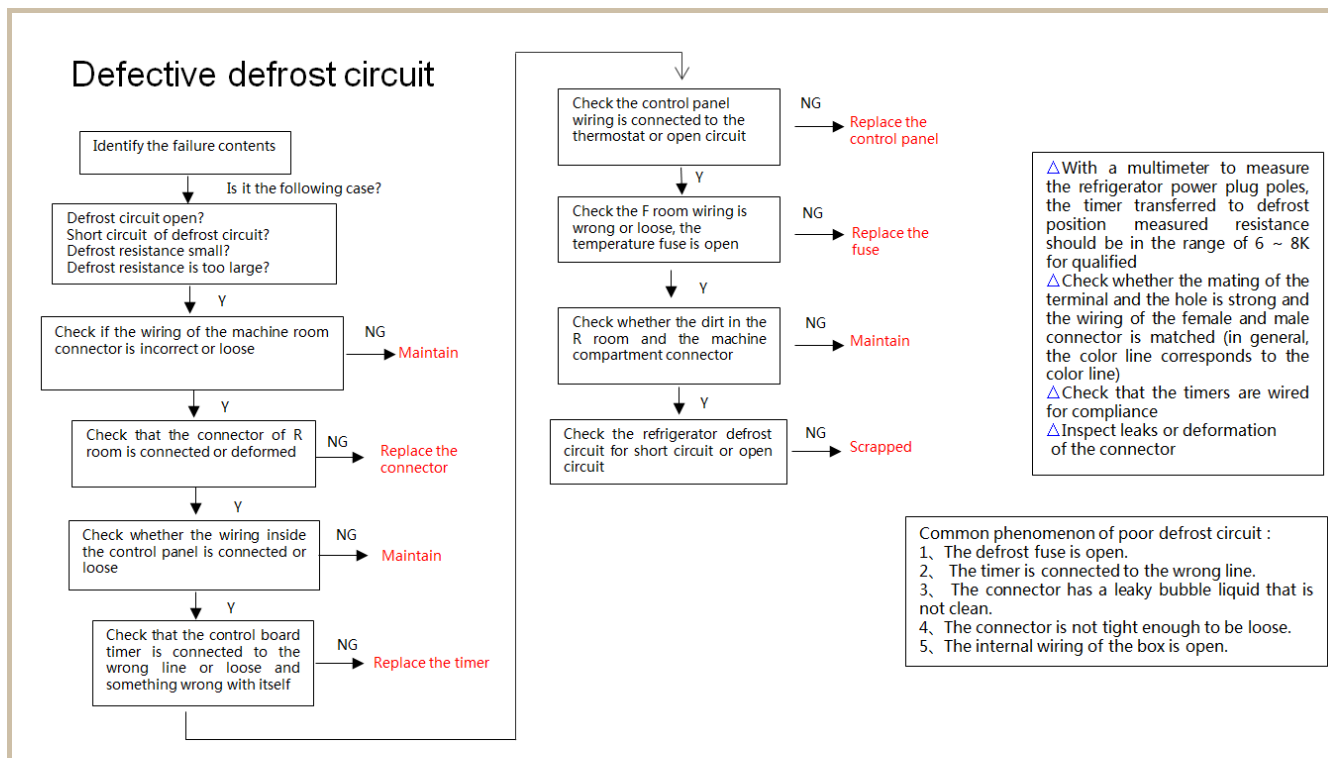
11.5 Light is not on



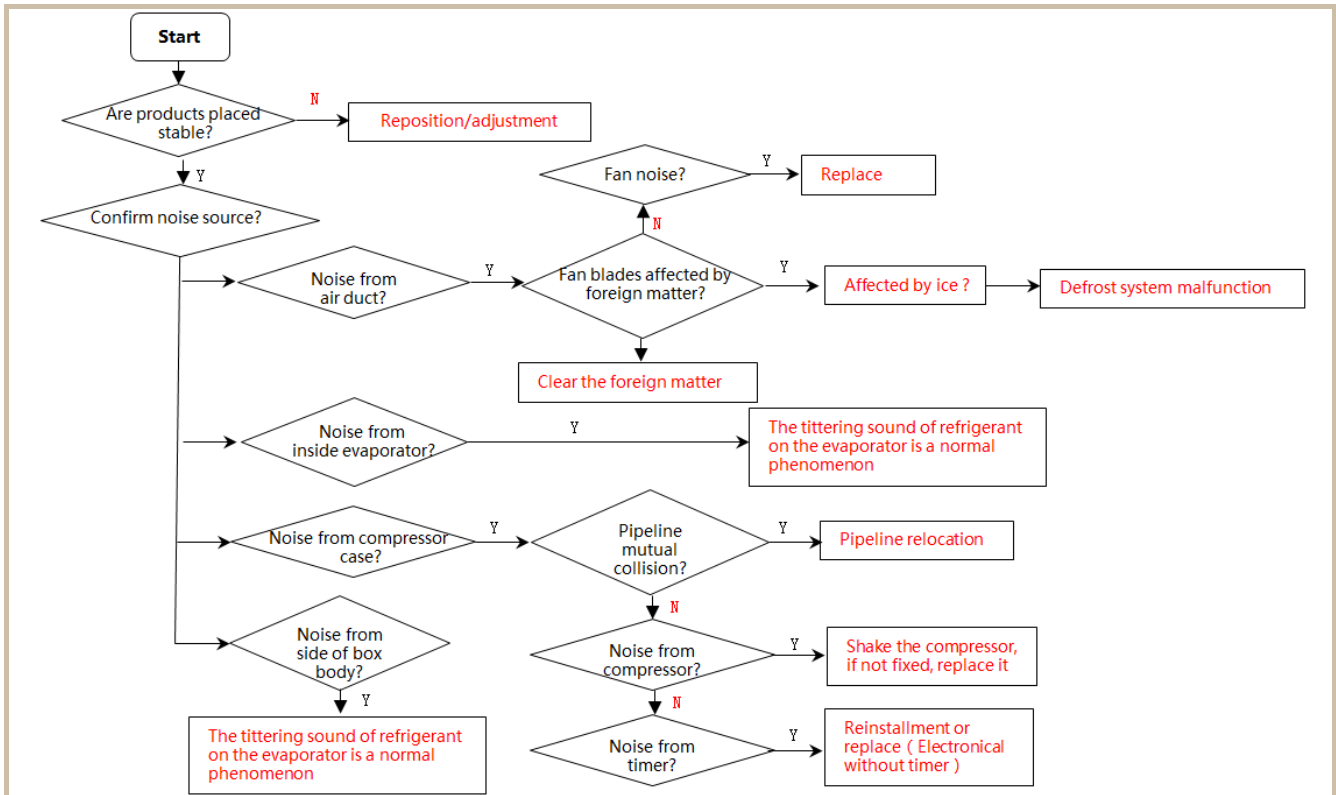
11.6 Fan failure



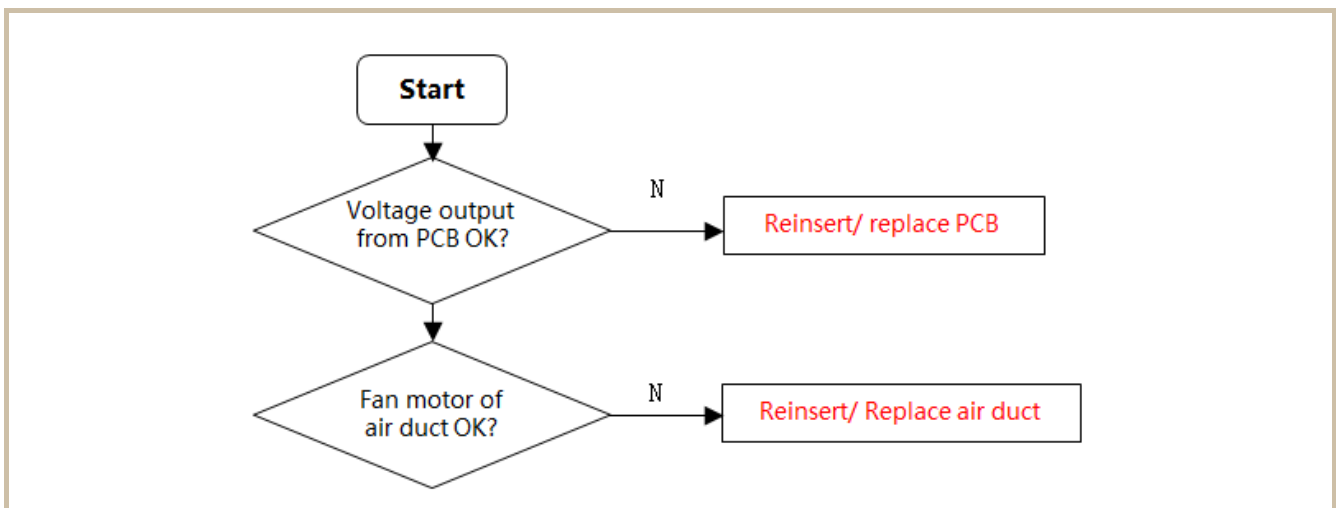
11.7 Defective defrost circuit



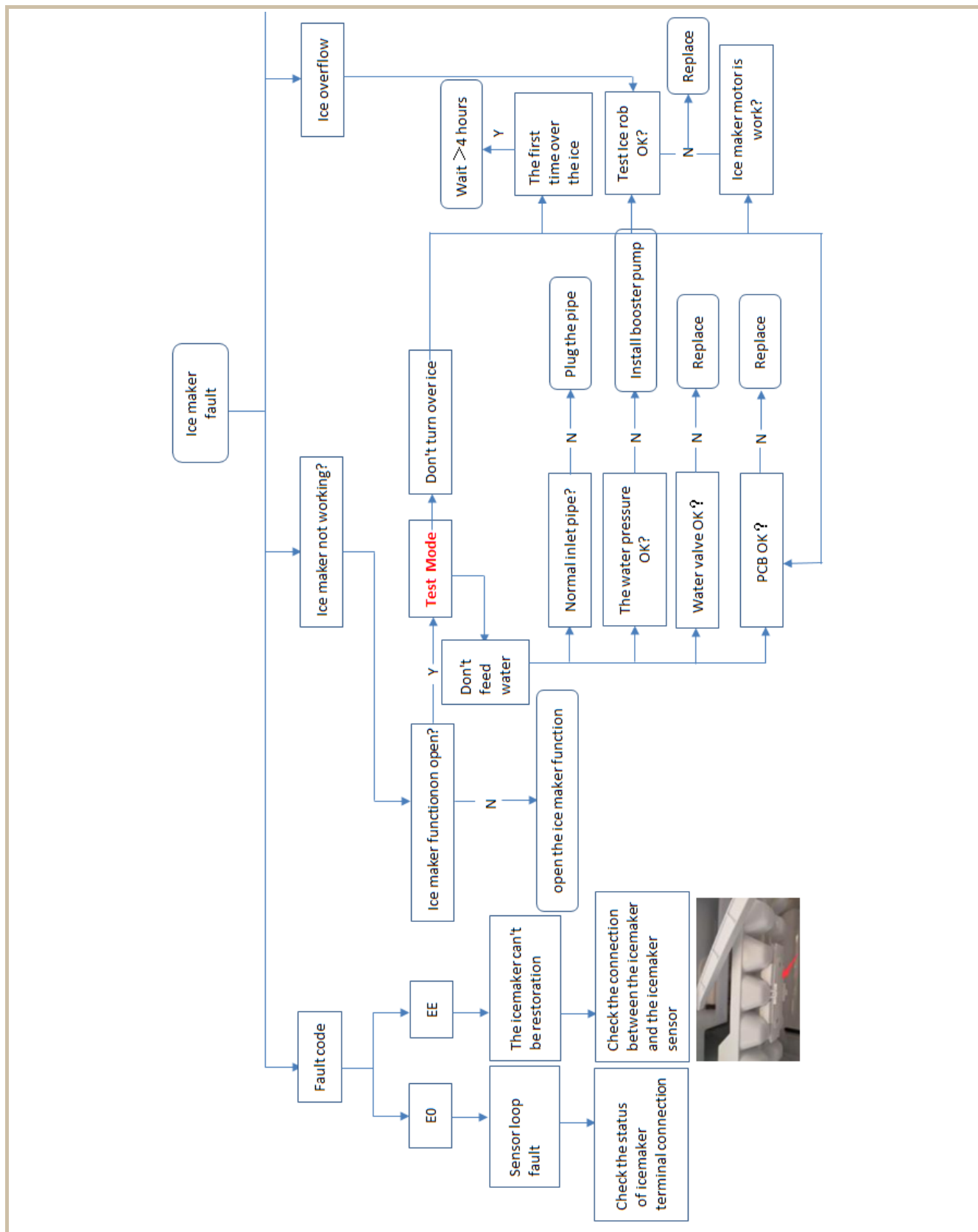
11.8 Noise



11.9 Air duct not operated (Option)



11.10 Ice maker not make ice (Option)





The symbol on the product or its packaging indicates that this product must not be disposed of with your other household waste. Instead, it is your responsibility to dispose of your waste equipment by handing it over to a designated collection point for the recycling of waste electrical and electronic equipment. The separate collection and recycling of your waste equipment at the time of disposal will help conserve natural resources and ensure that it is recycled in a manner that protects human health and the environment. For more information about where you can drop off your waste for recycling, please contact your local authority, or where you purchased your product.